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1 INTRODUCTION

In this report (MS3.3) we provide the interim view of the mechanisms and approaches developed in WP3 of EXIGENCE. It consolidates progress since Deliverable D3.1, integrates updates from MS3.1 and T3.1 discussions, and reflects architectural and metric dependencies from WP1 and WP2. The report's role is to capture the present status of carbon-aware orchestration, runtime service mechanisms, and incentive schemes, and to prepare the ground for Deliverable D3.2.

1.1 PURPOSE OF MS3.3

MS3.3 sits within WP3 as an interim milestone linking prior outcomes (D3.1, MS3.1) with ongoing development in T3.1–T3.4. Its purpose is threefold:

- Summarize and condense the progress achieved in the design and implementation of carbon-aware orchestration, runtime mechanisms, and incentives since D3.1.
- Document the integration points with WP1 architectural specifications and WP2 measurement/data frameworks, highlighting dependencies where work is still ongoing.
- Provide a coherent status update towards D3.2, ensuring alignment across WP3 tasks and maintaining consistency with overall project objectives (PO2 in particular).
- Consolidate the developments within WP3 on carbon-aware orchestration, runtime mechanisms and incentive schemes, ensuring their readiness for validation and demonstration activities in WP4, where ~~these mechanisms~~ candidate mechanisms will be integrated into the project testbeds to produce demonstratable outcomes at TRL4.

1.2 SCOPE

The scope of MS3.3 is to record the current state of WP3 mechanisms and policies, emphasizing their evolution, integration readiness, and demonstrable outcomes within the project timeframe. The report focuses on:

- Orchestration updates, including Kubernetes and MANO extensions for energy/carbon awareness.
- Runtime service mechanisms, including distributed scheduling and carbon-aware adaptation.
- Incentive frameworks under development, especially in user-centric use cases.
- System integration outcomes and cross-package linkages, while deferring unresolved inter-domain and multitenancy aspects to related deliverables (D2.2, D2.3, D3.2), where these topics are addressed in their respective technical scopes.

~~Each mechanism introduced in WP3 contributes to the PO2 target of reducing energy consumption and CO₂e by a factor of 3 for eMBB-type use cases and 5 for high-SLA PNI-NPN cases. Preliminary simulation and analytic assessments based on partners' datasets indicate:~~

- ~~• **Carbon-aware orchestration:** 20–30 % reduction in resource over-provisioning and up to 25 % lower total energy use under realistic traffic patterns.~~
- ~~• **Runtime service adaptation** (DRL-based scheduling): 10–15 % further improvement through load-adaptive placement and dynamic scaling.~~

- **Incentive mechanisms** (user and provider): 10–20 % behavioral reduction potential, depending on adoption rate and pricing elasticity.¶¶

Assuming partial independence between orchestration and runtime optimization layers, cumulative achievable efficiency gains approach 35–45 % in mixed scenarios consistent with progress toward the PO2 factors. Cross-validation in D3.2 and WP4 testbeds will confirm additive versus overlapping effects and guide prioritization for integrated deployment. Each mechanism introduced in WP3 contributes to the PO2 target of reducing energy consumption and CO₂e. The results from WP3, based on extensive empirical evaluations and published papers in top-tier conferences, confirm that the PO2 objectives are achievable. Specifically:¶¶

- **Carbon-aware orchestration:** Testing has shown a **20–30 % reduction in resource over-provisioning** and up to **25 % lower total energy consumption** under realistic traffic patterns, demonstrating the effectiveness of the optimization mechanisms.¶¶
- **Carbon-aware quality adaptation:** Evaluation results for AI inferencing tasks show that adjusting service quality in response to carbon intensity can reduce emissions by up to 10% beyond savings related to increased energy efficiency¶¶
- **Runtime service adaptation:** Initial evaluations indicate a **10–15 % improvement** in energy efficiency through adaptive placement and dynamic scaling techniques based on load variations, as verified in our experiments.¶¶
- **Incentive mechanisms (user and provider):** Real-world tests reveal a **10–20 % reduction** in energy usage through behavior-driven optimization, depending on the level of user adoption and pricing models.¶¶

Furthermore, recent results confirm that the other PO2 objectives are achievable:¶¶

- **Factor 3 savings** have been demonstrated in training models, both through recent research and ongoing work from project partners.¶¶
- **Factor 5 savings** in service provisioning are also supported by the ongoing work in WP3.¶¶

We also show in D3.1 that an AI training task can be performed using renewable energy only, bringing its CO₂e to zero. Additionally, D3.1 confirms that AI training tasks can be performed using renewable energy only, bringing their CO₂e emissions to zero, highlighting the feasibility of achieving net-zero emissions in specific scenarios.¶¶

These findings validate the expected progress toward PO2, confirming that the targeted energy and CO₂e reductions are well within reach.¶¶

¶¶

2 CARBON-AWARE ORCHESTRATION MECHANISMS

2.1 SUMMARY OF PROGRESS SINCE D3.1

Orchestration is the backbone of managing the complexity inherent in distributed Cloud Native (CN) systems based on microservices. Its mechanisms automatically manage the deployment, migration, and removal of workloads, basing their actions on established policies and increasingly sophisticated algorithms designed to achieve specific operational objectives.

A critical aspect of this process, especially in resource-constrained environments such as the edge, is careful resource management. It is essential to evaluate CPU, memory, network, and storage usage. In addition, the orchestrator must maintain knowledge of the infrastructure topology, as hardware availability is not uniform, and operational policies often introduce constraints beyond mere capacity, dictating when and which resources should be used. The shift from traditional virtual network functions (VNFs) to container-native cloud network functions (CNFs) or Kubernetes network functions (KNFs) underpins this evolution. While MANO continues to provide orchestration across diverse environments, including virtual machines, the ability to orchestrate containerized workloads is increasingly becoming the focus. At the same time, the main objective of green service scheduling is to improve coordination tools, such as Kubernetes and MANO, by integrating runtime algorithms that optimize resource efficiency, energy, and carbon emissions. Current advances focus on moving from basic resource-aware scheduling, which seeks greater efficiency through dynamic and adaptive resource management, to energy-efficient scheduling, which prioritizes minimizing consumption through the joint optimization of numerous technical parameters (such as task size and power), and finally to carbon-aware scheduling. This higher level of policy refinement has the explicit goal of reducing a service's carbon footprint by promoting the use of green energy resources, largely through techniques such as workload shifting. Since D3.1, these mechanisms have progressed from conceptual design to implementation-ready prototypes. The carbon-aware scheduling policies were embedded into the vEAS orchestration framework, extending Kubernetes and MANO with live carbon-intensity telemetry. Preliminary evaluations confirmed reduced resource over-provisioning ($\approx 20\%$) and improved node utilization under varying energy-mix conditions. The eco-control loop now dynamically adapts to workload profiles and geographic energy variability, forming a validated baseline for integration with WP4 demonstrators.

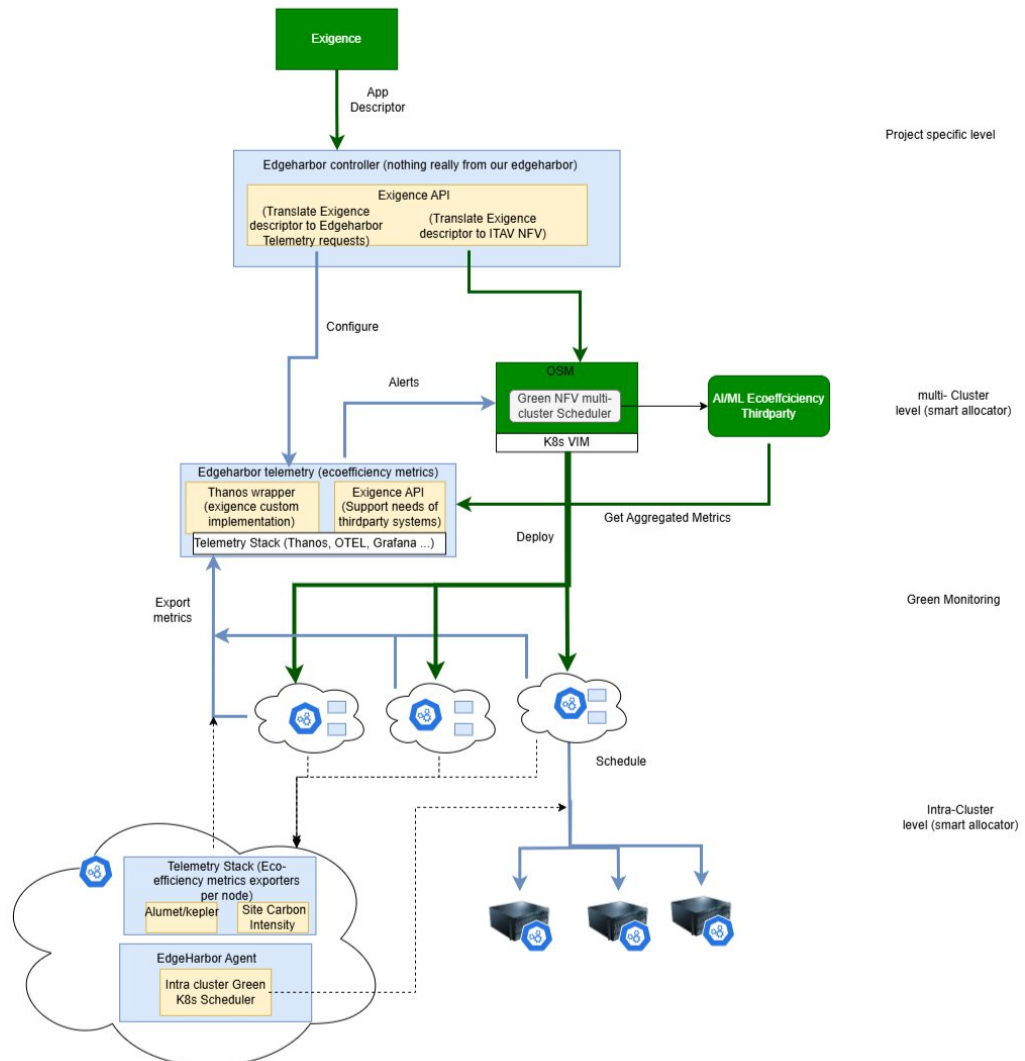
2.2 UPDATES IN STRATEGY AND IMPLEMENTATION

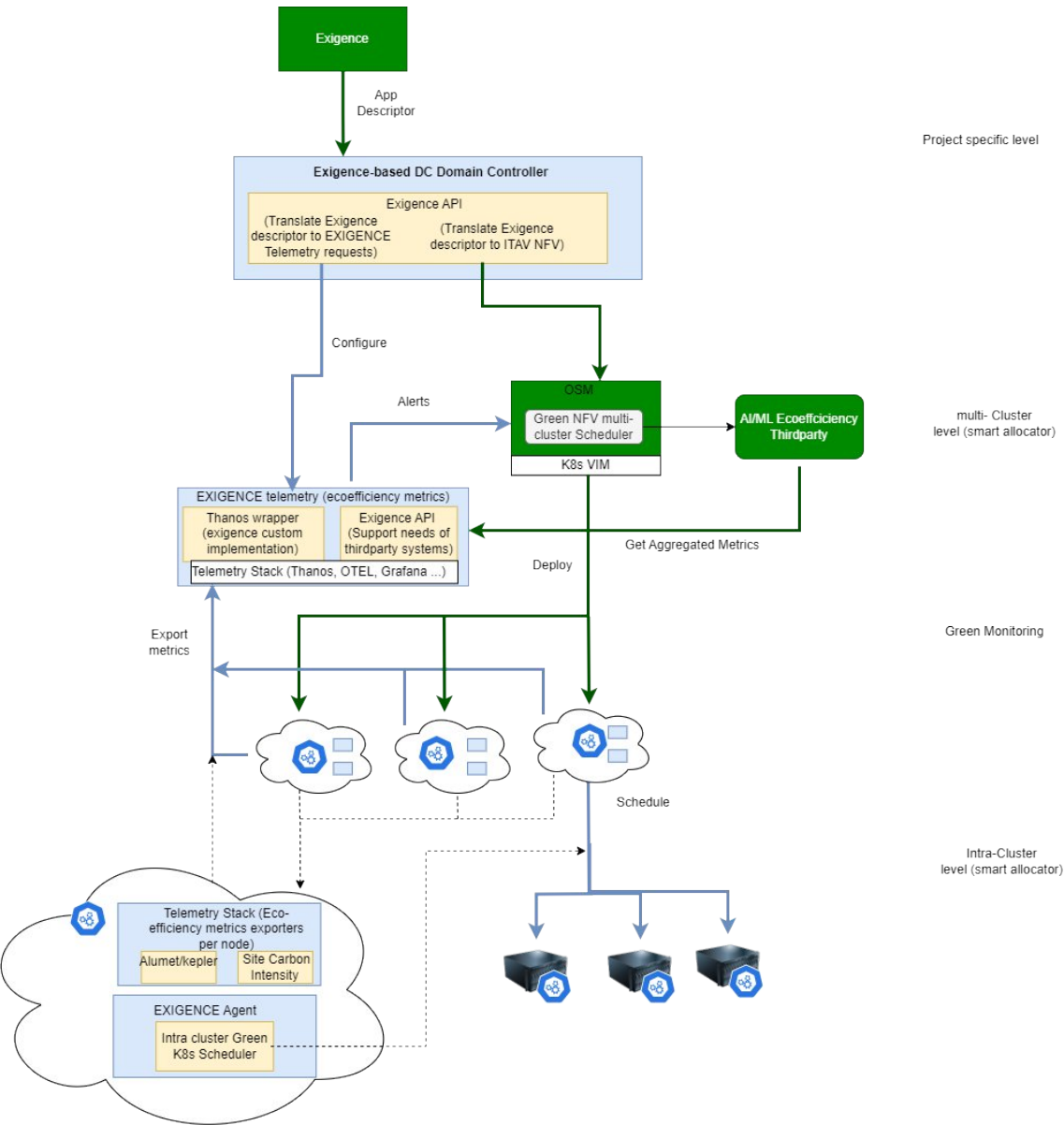
As a starting point, the initial integration of the different components that make up the EXIGENCE orchestrator (OSM integrated with K8s, acts as Vim) has been carried out, as well as the implementation of a first version of the K8s scheduler capable of assigning a score to nodes based on the carbon footprint they produce. At the same time, also the telemetry system that will provide this information to the orchestrator has been implemented. The next step has focused on deploying the telemetry system in the different clusters/nodes of the infrastructure to perform initial tests and validate its operation. The system exposes different endpoints that provide the information stored in the telemetry system, which is necessary to decide where to deploy applications/workloads, both in terms of energy consumption and carbon footprint.

This implementation aligns with WP2 definitions of carbon- and energy-related metrics, ensuring that the telemetry data used for orchestration reflects the same semantic constructs as those established in WP2. The orchestrator's optimization logic is thus based on validated, measurable indicators rather than abstract estimates. The resulting vEAS-Kubernetes integration is also prepared for inclusion within the WP4 demonstrator codebase for end-to-

end validation. In parallel, the EXIGENCE Agent interfaces with the domain-internal functional architecture defined in WP1 for the data-center domain, enabling bidirectional telemetry flow between the orchestration layer and local resource controllers. Together, these linkages ensure full continuity between measurement (WP2), orchestration and optimization (WP3), and validation (WP4).

Figure 1 shows a high-level graphical representation of the orchestration system architecture.





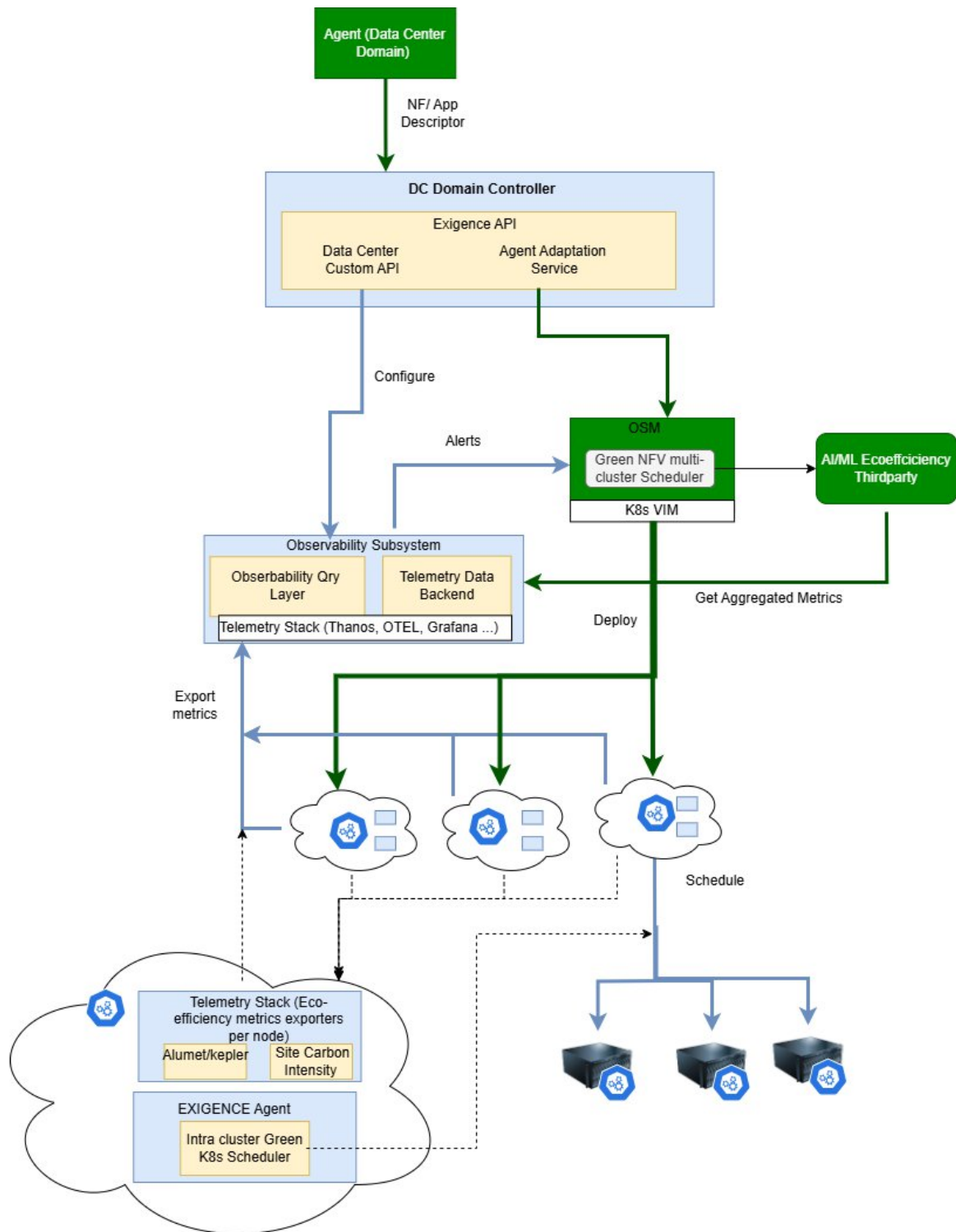


Figure 1 EXIGENCE orchestration system high-level architecture

2.3 INTEGRATION WITH EXISTING PLATFORMS (E.G., KUBERNETES, MANO) – (EVIDEN,ITAV)

Open Source MANO (OSM) is the MANO platform used within WP3 for the orchestration framework, deployed in the project's virtualized testbed environment and extended to incorporate energy- and carbon-aware orchestration mechanisms, extended to incorporate energy- and carbon-aware orchestration mechanisms. Two different mechanisms are added to OSM: a service throttling mechanism and a sustainable placement mechanism.

The service throttling mechanism introduces virtual Energy-Aware States (vEAS), representing different operational states a VNF/CNF can operate at, with specific resource configuration corresponding to the desired performance/energy consumption expectations. These vEAS are defined in ETSI ES 203 682, and the OSM VNF and NS Information Models (IMs) are extended to include the fields. [These new fields follow the semantic definitions of the energy and carbon metrics established in WP2, ensuring that the orchestration layer operates on consistent measurement concepts. These extensions have been communicated to WP2 so that the same metric structures can be reflected in their deliverables and ongoing work on telemetry data models. The OSM NS instantiation and update endpoints are extended to include vEAS identification, allowing for dynamic changes of the NS operational state.](#)

The service placement mechanism extends the current Network Service (NS) instantiation process by enabling OSM to make NS placement decisions that follow sustainability-driven criteria, deploying the NS in the most sustainable manner possible. A new NS instantiation endpoint is created that, instead of having the user define a VIM for the NS to be deployed, does not support any VIM specification, leaving the decision of which VIM to deploy the NS to be made by OSM, which will decide the most sustainable VIM based on its environmental impact profile. The metrics used for this decision making are the Power Usage Effectiveness (PUE), which describes the energy efficiency of a datacenter infrastructure, and Carbon Usage Effectiveness (CUE), which describes its carbon footprint. [Both metrics are consistent with the energy and carbon indicators defined in WP2, ensuring semantic alignment across measurement and orchestration layers.](#)

To enable both these mechanisms, besides the extensions made to OSM NBI and IM, a new function named Eco-control was introduced, deployed in the OSM Kubernetes cluster in the same namespace. The Eco-Control module records the vEASs of each NS managed by OSM, and coordinates state transitions by interacting directly with the VIM to enforce the NS state resource configuration changes. The Eco-Control module also registers the Energy and Carbon profile of the VIMs it oversees, enabling real-time sustainability-aware NS placement across them.

With regard to Kubernetes, the process used for placement/scheduling workloads on different nodes within a cluster is designed as an extensible framework based on APIs and plugins, shown in Figure 2.

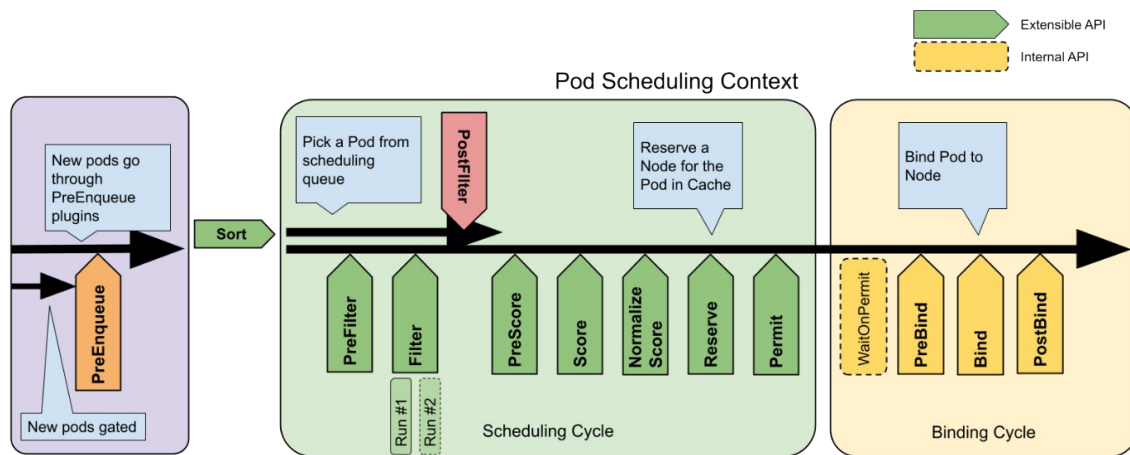


Figure 2 Kubernetes Scheduling Framework extension points. (source: <https://kubernetes.io/>)

Considering the set of extensible APIs shown in Figure 2, the internal mechanics can be reviewed and studied where a potential plugin for eco-efficiency fits better into the process. Here is how pods/workloads are scheduled:

First, there is a filtering phase in which nodes that cannot host the pod/workload are discarded based on basic rule checks, such as whether the node has sufficient CPU and memory, whether it can tolerate the pod's taints, and whether the necessary host ports are free.

Once the filter has reduced the list of nodes, the scheduler enters the scoring phase. The nodes that have passed the filter are evaluated, assigning a score to each of them, with a higher value for the best candidates. To do this, the scheduler runs all the scoring plugins, including those that have been created to extend the scheduler, on each node. In this process, each add-on returns its own score, which is then multiplied by a weight to determine its importance. This allows the operator to control which scoring factors are most critical, such as affinity or resource utilisation.

Finally, the selection is made by adding up all the weighted scores. To do this, the scheduler chooses the one with the highest score. If there is a tie, one is selected at random. Once the decision is made, the scheduler updates the pod, “binds” it to the chosen node, and the kubelet on that node takes control, starting the container.

For Exigence, it has been decided to implement a score plugin to include eco-efficiency metrics into the node selection process in a cluster and thus evolve the default k8s scheduler into a K8s Green scheduler. The proposed plugin leverages the monitoring framework of EXIGENCE and the curated set of metrics related to carbon emissions to prepare a green score for each node based on the software carbon intensity (SCI) and then rank them to favor the lowest SCI value.

In addition to the score plugin, carbon and energy metrics have also been made available to the K8s custom metrics API to enable the configuration of Horizontal Pod Autoscaler (HPA) for user-defined workloads based on the energy consumption or carbon footprint of a pod. The process consists of the deployment of a [prometheus adapter](#) in the cluster which connects to the EXIGENCE observability framework, reads an HPA configuration attached to a workload and pulls the associated metric to feed it to the custom metrics API of K8s. As a result, users can define autoscaling policies based on advanced metrics.

Currently, OSM does not support different operational states for its Network Services, which prevents the creation of service variants (different configurations of a service yielding distinct energy consumption levels), as proposed in D3.1. Additionally, it does not consider sustainability-driven criteria in the lifecycle management of Network Services, such as during instantiation, thereby limiting energy- and carbon-aware system scheduling, also proposed in D3.1.

The introduction of virtual energy-aware states allows the definition of different operational states for Network Services, each with its own energy consumption expectation, enabling the creation of network service variants. Moreover, by characterizing VIMs according to their energy/carbon profile, OSM can leverage that information to instantiate/migrate Network Services to more sustainable infrastructure, enabling energy- and carbon-aware scheduling.

By adding/extending the current OSM interfaces to expose these new “green” capabilities, we enable user empowerment (also proposed in D3.1), making the user aware of its environmental impact and the actions he can take to potentially reduce it. [These new OSM interfaces and green capabilities are being aligned with the functional architecture work in WP1. In the architectural model, OSM acts as the orchestration element capable of producing service variants based on environmental context, while the EXIGENCE Agent mediates between the orchestrator and each domain’s internal controllers. This linkage enables bidirectional information flow: the orchestrator can request Energy Consumption \(EC\) and Carbon Emission \(CE\) indicators from each domain through its agent, and domains can in turn act as service users when invoking external resources. The internal optimization procedure can thus compare the eco-cost of local versus external service execution, allowing decisions such as prioritizing local processing when greener or offloading to external services when more sustainable overall.](#)

Table 1 provides update on the requirements for the mechanisms proposed for further consideration in WP3 (table 4 in D3.1).

Table 1 Updated Summary of the requirements of the proposed mechanisms for Green Orchestration.

	GO: Resource Demand Matching	GO: Service Variant Creation	Intra-/Inter-domain applicability
Input	Service descriptor (Definition of the service, including VNFs/CNFs and their requirements). SLA Requirements (Quality of service (QoS) and latency constraints). Policy Rules (Energy efficiency, carbon footprint limits). Energy situation of the resources and their carbon footprint (fixed or dynamic, depending on the use case).	Per service, min. 2, typically 3-4 different configurations, known to have a different ecological impact, e.g. lower energy consumption (at the costs of quality), lower carbon emission (at the costs of quality), etc. Such different configurations are to be developed and benchmarked in advance. As potential generic service variant types we can think of modes for each service like these: 1. Energy-Saving Mode 2. Balanced Mode 3. Performance Mode	[Intra-domain] VNF/CNF Service Descriptors have a fundamental impact on the data centre domain, as they explicitly define the requirements of the VNF/CNF on the Network Function Virtualisation Infrastructure (NFVI). With regards to SLA Requirements in Data Center domain, the NFV Orchestrator (NFVO) is the key point where Network Service (NS) SLAs are defined (although customer define SLA through BSS) and fulfilled. It is responsible for taking SLA requirements and

	Resource situation: current known available resources/capacities and allocation situation. Optional: per-event or periodic updates on resource availability changes (if dynamically changing).	4. Carbon-Aware Mode Service descriptor (Definition of the service, including VNFs/CNFs and their configurations and requirements). Optional: SLA Requirements for each variant, which are not the same as SLAs for the main service, i.e. with specifically diverging degrees of SLA enforcement for each configuration, e.g. clarifying lower quality, relaxed availability, etc. Policy Rules (Energy efficiency, carbon footprint limits).	translating them into concrete actions on the infrastructure. Policy rules act as external constraints that guide the decisions of the MANO Orchestrator.
Algorithm	Optimization (Optimizing resource allocation and service placement). Typically ILP, MIP or GPM. Complex, centralistic, non-scalable (1000-10000 parameters). Machine Learning (Predicting power usage and carbon footprint).	Optimization (Optimizing resource allocation and service placement) Standard Lifecycle management of classical orchestration (deploy, maintain, adjust to load, remove at the end of lifecycle). Related but not part of Service Variant Creation: each variant can be later considered for GO/Resource Demand Matching optimization on its own, within the constraints pertaining to this variant. Machine Learning (Predicting power usage and carbon footprint).	[Intra and Inter-domain] In Data Center domain, optimisation algorithms usually reside and are executed by the MANO (NFVO), but their objective is to intelligently handle and configure resources in the NFVI/Data Centre domain and connectivity routes in the Network domain. There may also be an additional external layer that deals with AI.
Output	Deployment plan (Optimized placement and scheduling of services).	Initiation of all considered service variants.	[Intra-domain] The deployment plan is generated by MANO, and the NFVI/Data Center and Network domain must follow to ensure that the VNF is instantiated with the optimal configuration of performance, cost, and sustainability, previously calculated.
Compute	High compute requirements for the solver, e.g. a dedicated powerful server. High CPU Usage (Processing power needed for	The lifecycle management function per se is low on demands. Lookup for a recipe per service variant, deployment, publication in some service registry.	[Intra-domain] Orchestration (MANO) and Data Center (NFVI) domains.

	orchestration algorithms). GPU Utilization (If ML models are involved, GPUs may accelerate computations).		
Storage	Database-like abstraction with: Current allocation situation (for recovery after restart, for example). Policies, rights. Service descriptors. Historical data for analysis.	Negligible, historical data for analysis.	[Intra-domain] Orchestration (MANO) and Data Center (NFVI) domains.
Networking	Low-latency, reliable communication between orchestrator components for update fetching and command distribution from the orchestrator to the resource controllers and/or resources.	Negligible (Low-latency communication between orchestrator components).	[Intra and Inter-domain] Orchestration (MANO) and Data Center (NFVI) domains, and the Network domain (Control Plane Network) to ensure the required operational speed and reliability.
Energy	Execution: Considerable, for large-scale problems. AI-based solutions: High for the initial training of suitable ML models.	Execution: Low, given the low computation, communication, and networking requirements.	[Intra and Inter-domain] Orchestration (MANO) and Data Center (NFVI) domains.

The following table presents newer platform-specific mapping of WP3 requirements, including recent orchestration capabilities such as Kubernetes CRDs, MANO energy-aware states, and WP2 metric interfacing assumptions.

Table 2 Platform-specific mapping of WP3 requirements

Req.	Description	K8s Support	MANO Support	WP2 Data Used	Notes
GO.1	Dynamic scaling based on energy metrics	Partial (via CRDs)	Yes (via VEAS)	EC#3, EC#6	Orchestration mechanisms require a specific time resolution of requested metrics' ecodata. This time resolution is inherently limited by the capabilities of the metering tools and the underlying measurement methods they employ. As a result, different metrics may exhibit different granularities. To ensure consistency and reliability in the orchestration process, the orchestrator can retrieve the time resolution associated with each metric by inspecting the ecodata

					exchanged through the data exchange agents, i.e., each metric's value reported is accompanied with its sampling interval among other data provided.
GO.2	Agent coordination for multi-tenant workloads	Plugin-based	In roadmap	EC#7, CF#3	Multi-tenancy logic out of scope for WP3 While (domain) agents receive high-level intents from their coordinator/high-level orchestrator, they can also leverage ecodata collected from the observation and data collection layer, both within their own domain and across other domains. By analyzing this data, domain agents may potentially make (improved) autonomous orchestration decisions (if allowed by the policy). With such hybrid coordination model, overall system efficiency may be further improved.

2.4 SCALABILITY AND OVERHEAD ANALYSIS OF ORCHESTRATION MECHANISMS

Integrating energy and carbon awareness into Network Function Virtualization (NFV) orchestration introduces significant challenges regarding scalability and overhead, because it involves balancing sustainability, delivering appropriate performance, and dealing with computational complexity. The main challenge of scalability is that the problem of deciding where to place and when to activate VNFs/CNFs (placement and scheduling) significantly increases when a dynamic variable such as the carbon intensity of the electricity grid, which changes over time every moment, is added. This is currently a very complex problem that is computationally unfeasible for large Telco networks without using heuristics or Machine Learning (ML) models, even if they are not perfect, just to maintain operational speed.

Achieving this balance between service quality and sustainability introduces an overload in two ways. First, there is the control plane overload, as for orchestration to work optimally, the NFVI (data center) and MANO (orchestrator) must be in constant communication. They need a continuous flow of super-granular telemetry data on the energy consumption and carbon footprint of the electrical network. Based on the data received, the orchestrator has to send new commands (for relocation and scaling) quickly, which requires low-latency and high-reliability communication.

At the same time, the continuous execution (inference) of models for workload prediction, although not massive, adds a considerable continuous energy load to NFVI servers. But the really significant expense is from the initial training of those models, which consumes an extremely high amount of energy.

Achieving this balance between service quality and sustainability introduces an additional control-plane load, as NFVI (data center) and MANO (orchestrator) must exchange telemetry to maintain optimal operation. However, continuous data streaming is not required. The system adopts an event-driven model, transmitting updates only when monitored parameters (e.g., energy consumption, carbon intensity, or resource utilization) change beyond predefined thresholds. This approach minimizes control-plane traffic while ensuring that orchestration reacts promptly to significant environmental or operational variations. Currently, the most successful and mature implementations are intra-domain. This means that carbon optimisation occurs within a single data center or a single operator's network, as the orchestrator has full visibility and control over internal resources and energy metrics.

Cross-domain orchestration, involving multiple authoritative domains or operators, presents well-known challenges such as lack of standardized carbon-accounting models and data-sharing constraints. While these issues are recognized as relevant to achieving fully end-to-end sustainable services, they are beyond the current scope of EXIGENCE, which focuses on intra-domain orchestration and optimization within trusted environments. The mechanisms developed here, however, are designed to remain compatible with future extensions toward federated or multi-domain scenarios. However, the main challenge, the task at hand, is cross-domain orchestration. To offer truly green and sustainable end-to-end services (e.g., between multiple operators or using federated clouds), the challenge here is that there is no single standard for measuring carbon footprint, which leads to biased decision-making by algorithms.

Another major problem is data privacy, as operators do not usually share their network topology or exact energy consumption with an external coordinator.

3 CARBON-AWARE RUNTIME SERVICE MECHANISMS ~~(HWDU)~~

3.1 UPDATED MECHANISM DESIGNS

3.1.1 CARBON-AWARE QUALITY ADAPTATION FOR ENERGY-INTENSIVE SERVICES

The energy demand of modern cloud services, particularly those related to generative AI, is increasing at an unprecedented pace. To date, carbon-aware computing strategies have primarily focused on batch process scheduling or geo-distributed load balancing. However, such approaches are not applicable to services that require constant availability at specific locations due to latency, privacy, data, or infrastructure constraints. In [22], we focus on the temporal variability of carbon intensity and explain how adapting service quality with respect to this metric does not only impact energy demand but can yield additional carbon savings. We define the quality of responses (QoR) as a type of quality of service (QoS) metric that describes the proportion of requests served by two different service quality tiers. For instance,

the service quality and carbon footprint of LLM inference is determined by many factors such as the choice of model, quantization and pruning, and the number of generated tokens.

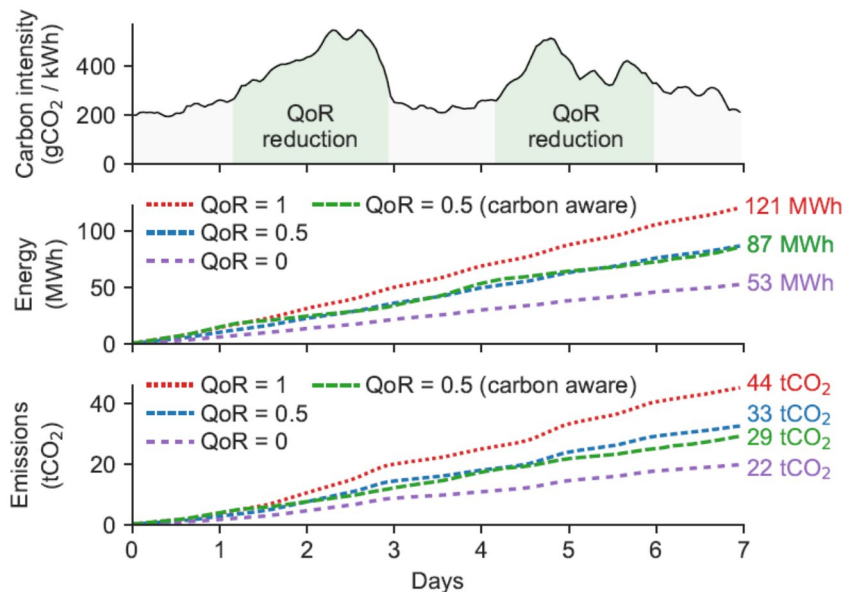


Figure 3: Adjusting the quality of responses (QoR) with respect to carbon intensity can reduce emissions beyond savings related to reduced energy demand (from 33 tCO₂ to 29 tCO₂ in 7 days).

Figure 3 shows the accumulated energy usage and carbon emissions for an example scenario with two quality tiers over one week. Here, serving all requests at the high-quality tier (QoR = 1) uses roughly twice the energy compared to serving them at the low-quality tier (QoR = 0). Balancing requests evenly between the two tiers results in intermediate energy usage. We explore how strategically timing quality reductions based on carbon intensity can further reduce emissions by performing a comprehensive analysis across different regions, request patterns, and QoR constraints.

We formalize the problem of minimizing emissions under QoR constraints through optimized resource provisioning and load balancing between two service quality tiers. We show that the optimal solution is NP-hard ~~complete~~ and requires perfect knowledge about the future carbon intensity and traffic. We propose an online forecast-based multi-horizon (long-term and short-term) optimization for minimizing emissions that reaches close-to-optimal carbon savings. We evaluate the practicability of carbon-aware QoR adaptation by applying our online approach under real forecasts and approximated MILP solutions: Long-term optimizations are stopped after 30s; short-term optimizations after 10s.

Our results demonstrate that adjusting service quality in response to carbon intensity can reduce emissions by up to 10% beyond savings related to increased energy efficiency. We present the results for different regions and different user traffic patterns as shown in Figure 4. As is typical in the assessment of QoS metrics, QoR is defined over validity periods of fixed length. These periods can either be disjointed (e.g., quarters, months, weeks) or, more commonly, overlapping. We evaluate on ten globally distributed regions: California, US (CISO), Texas, US (ERCOT), New York, US (NYISO), Germany (DE), Spain (ES), Netherlands (NL), Poland (PL), Sweden (SE), and Queensland, Australia (AU-QLD). All carbon intensity data was provided by ElectricityMaps [15].



Figure 4: Relative (top) and absolute (bottom) additional carbon savings for different validity periods of our proposed online approach (colored) compared to upper bound potential (Gray). Larger validity periods enable higher savings.

To the significant potential for carbon efficiency improvements in cloud services, ChatGPT, one of the most energy-intensive cloud services today, reached over 200 million active users in 2024 [16]. Assuming just 5 requests per user per day, this amounts to nearly 42 million requests per hour on average. Assuming performance similar to Llama 3.1 70B [17], we estimate that the annual operational carbon emissions of large-scale LLM service providers currently exceed multiple 10,000 tCO₂—a figure set to rise with multi-modal AI models.

3.1.2 SERVICE AND NETWORK FUNCTION DISTRIBUTION MECHANISMS

Edge Computing enables the deployment of services closer to the end users, which reduces the communication latency, offloads the core, and allows more efficient resource management and contains potential fault scenarios to local areas only directly affected by the fault. Indeed, services that are traditionally deployed on centralized servers can now be distributed over multiple Edge servers, and the resources allocated to them can be adapted on-demand to adjust to local requirements. The User Plane Function (UPF) is one candidate for distributed Edge deployment, where multiple lightweight UPF instances can be deployed on Edge servers with sufficient resources to accommodate the current users and load. This architecture, however, also introduces potential fault scenarios that may affect specific local areas directly, requiring appropriate fault-tolerance and recovery mechanisms to ensure service continuity.

In this setting, it would be very beneficial for an operator to use the same compute infrastructure for network functions and hosting services whenever possible. Having a single platform for both network functions and services improves energy efficiency by consolidating resources and reducing duplication. A unified cloud enables shared compute and cooling systems, which minimizes idle capacity and optimizes utilization. This integration also allows for dynamic resource allocation, so workloads can scale up or down based on demand without overprovisioning hardware. Additionally, centralized management in a single cloud environment supports advanced energy-saving techniques like workload placement based on power consumption and leveraging renewable energy sources more effectively. Overall, this

approach reduces hardware footprint, lowers power usage, and streamlines operations, leading to significant energy savings. Hence~~However~~, NFs would compete for resources with service processing functions. Although this provides benefits in terms of physical infrastructure costs, it presents a big challenge on the sharing of these resources and their allocation to specific functionalities, as well as the tradeoffs involved. ~~However, t~~This issue has not been addressed since they are usually seen as separate problem spaces, and NFs are currently envisioned to run on specific private telco clouds. However, this issue has not been addressed since they are usually seen as separate problem spaces, and NFs are currently envisioned to run on specific private telco clouds. This separation will make operating mobile networks of the future more expensive.

~~This separation will make operating the wide area networks of the future more expensive.~~

The main source of the problem with a common infrastructure for NFs and service functions is that the impact they have on each other is not well researched and, so far, poorly understood. In terms of resources, they use the same resources present on computing infrastructures – CPU, Memory and Networking. However, they need different portions of them at different times; so, when both types draw from the same resource pools, contention issues may arise and decrease performance. User Plane NFs (UPFs in 5G) create most of their load on the network cards and some of it on the CPU and Memory via the PCI bus on a machine, while service processing functions create most of their load on CPU and Memory while some of it is on the network for incoming and outgoing traffic. Since I/O operations for storage are negligible in these cases, we do not consider them at this stage. Although this may seem as a (mostly) disjoint resource problem, during intensive resource sharing near resource pool limits it can be very difficult to estimate what the impact of this resource sharing is. Furthermore, distributed locations with a central management point (i.e. cloud management) allow for mixing and matching deployments and resource allocations with many options.

3.1.2.1 SYSTEM MODEL AND ALGORITHM

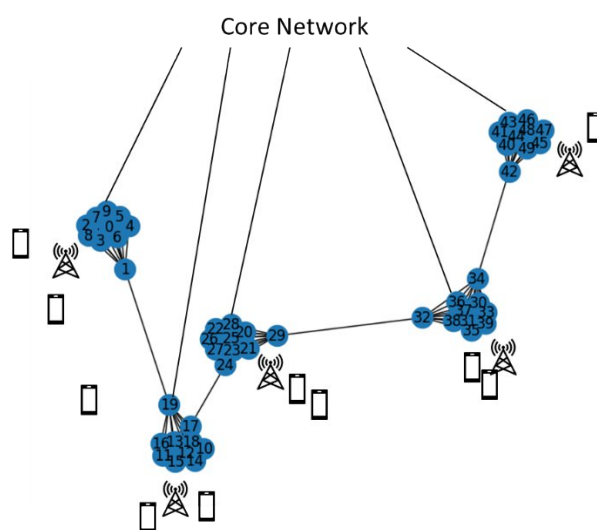


Figure 5 Computing and Network Infrastructure

To deliver the service described in D1.2 as UC14, there is an underlying computing infrastructure with multiple edge data centers in the vicinity of the event and multiple wireless access points. This is illustrated in the figure above, where each graph node represents one

compute machine that can host NFs or SFs. We assume that the hardware resources per machine in the edges are fixed (CPU and network), and shared among the deployed network, and service functions (SFs) and applications which is implemented by the resource scheduling algorithms in the Operating System of each machine. The resources are represented by Queueing Theory models, with the main performance metric being latency. We focus on the idle energy needed for a machine to be powered on whenever there is an NF or SF that cannot be moved to other machines.

In this setting, the goal is to minimize the overall energy use while keeping the latencies within SLAs. Currently, we are solving small scale problems with Mixed Integer Non-Linear Programming (MINLP) to establish a baseline for RL-based solutions that scale to large problems. An environment representing this is also in development with the Python module “gymnasium”, after which we will develop the algorithm. These challenges will be addressed in the next phase, and the proposed solution would be reported in D3.2.

3.2 SUITABILITY FOR DISTRIBUTED EXECUTION

For the solutions proposed in D3.1, Sec 7.1 – 7.3, and in Section 3.1.1 of this report, we require to distribute and execute instances of the algorithms on nodes (e.g., the nodes running the service) or computing centers (e.g., on the ingress node). Nodes are using estimation of the load-energy profile and the traffic load at next service-stage (or endpoint) nodes to decide immediate redirection, next hop selection, or service QoS level. Such information can be provided to nodes using a distributed data layer or distributed techniques such as gossiping. These solutions can be therefore scaled to large networks. Table 3 summarizes the requirements for the proposed solution.

Table 3: Summary of the requirements for the mechanisms proposed in D3.1 Sec 7.1-7.3 and Sec 3.1.1 of the current report.

	RSS: System Scheduler	RSS: Service Scheduler
Input	D3.1 – Sec. 7.1 Several next hops suitable for object/flow/request Load-energy-profile + est. current load on these hops (obtained async, e.g., through gossiping, refreshed when available) Communication quality (rate and latency est.) to these hops Current carbon intensity on at least some of these hops	D3.1 – Sec 7.2 & 7.3 Several next service-stages (or endpoints) nodes, suitable for object/flow/request Load-energy-profile + est. current load on these hops (obtained async, refreshed when available) Communication quality (rate and latency est.) to these hops Current carbon intensity on these hops Sec. 3.1.1 Load-energy-profile + est. current load on local servers (available locally) short-term and long-term carbon-intensity and work-load forecasts
Algorithm	RL-based, trained offline and fine-tuned and applied online, inference	D3.1 – Sec 7.2 RL-based with execution complexity of $O(1)$.

	complexity of $O(1)$. Infrastructure (control plane) is required to distribute execute the scheduler codes close to where the service requests occur.	D3.1 – Sec 7.3 Binary selection, av. and worse case $O(\log(K))$ D3.1 – Sec 7.2 & 7.3 Infrastructure is required to distribute and executed the scheduler codes on the nodes, e.g. the servers and the clients, running the services. Sec. 3.1.1 An approximative MILP solution with execution complexity $O(1)$. The infrastructure needs to run instances of the scheduler codes at each computing site.
Output	Immediate redirection or next hop selection decision	D3.1 – Sec 7.2 & 7.3 Immediate redirection or next hop selection decision Sec. 3.1.1 QoS level and the node to run the service.
Compute	Low, but might require local processing capabilities, including programmability of some entities to execute a particular scheduling code	D3.1 – Sec 7.2 & 7.3 Low, but might require particular service clients, to be able to select next-hop according to known service-energy parameters. Sec 3.1.1 Low.
Storage	Negligible, status information	Negligible, status information
Networking	Negligible (Async. update of next hops plus status information, sending selection and redirection commands). Status information update may occur more frequently, compared to that for orchestration, but such updates are required for much smaller set of nodes.	D3.1 – Sec 7.2 & 7.3 Negligible (Async. update of next hops plus status information, sending selection and redirection commands). Frequent status update for the (relatively small set of) nodes involved in the service execution. Sec 3.1.1 Negligible. Status update at servers running in a cite.
Energy	Execution: Low, given the low computation, communication, and networking requirements High training cost, but the trained model has good generalization and transfer-learning properties, thus can be everywhere: low marginal cost.	D3.1 Sec 7.2 AI-based solutions: High training cost, but can be applied everywhere, thus low marginal cost. Execution: Low, given the low computation, communication, and networking requirements Sec 3.1.1 & D3.1 Sec 7.3 Execution: Low, given the low computation, communication, and networking requirements

3.3 METRICS AND INTERFACING WITH DATA SOURCES

Table 4 summarizes the required energy and carbon footprint consumption metrics to run the proposed scheduling solutions, outlining the dependencies between these runtime scheduling decisions and the metric provided by the WP2 system:

Table 4: Summary of the required energy and carbon footprint consumption metrics by the proposed mechanism

Metric	Source (WP2)	Consumer (WP3 mech.)	Required Granularity	Update Frequency	Note
EC#2/3/6	EC of HW/SW /VE	Service schedulers in D3.1 Sec 7.2 & 7.3, and in Sec 3.1.1	Per session (a training round an inference session) and per node.	Per training rounds (usually in order of few minutes) or every hour (algorithm of Sec 3.1.1)	Needed for server selection and QoS level determination.
EC#8	Renewable Energy	Same as above	per node/HW, or per site (if shared among multiple nodes)	Every few minutes (e.g., every 5 minutes)	Same as above
EC#10	Load-profile	Same as above	Per node/HW	Ideally the load profile is measured offline, so no update is required.	Same as above
CF#2/3	CF of using a HW/SW	Service schedulers in D3.1 Sec 7.3 and in Sec 3.1.1	Per session and per node	Per training rounds (usually in order of few minutes) or every hour (algorithm of Sec 3.1.1)	Needed for server selection and QoS level determination.

3.4 COMPLIANCE AND SCALABILITY OF RUNTIME MECHANISMS

In this section, we focus on a single domain, and on how to schedule services over resources to improve the energy- and carbon-efficiency of the system. We observe that our solutions are scalable and rely on energy measurement metrics which have been already proposed in WP2, thus current assumptions and measurements in WP2 suffice. Besides, these solutions have low execution complexity and thus can be performed in runtime. The integration of these scheduling mechanisms within the Exigence architecture is currently being investigated and would be reported in D3.2.

4 INCENTIVE MECHANISMS FOR CARBON-AWARE BEHAVIOR

The aim of this section is to present novel incentive mechanisms to motivate energy-aware efficient consumption of ICT services.

4.1 SUMMARY OF MECHANISMS: ECONOMIC, BEHAVIORAL, AND REGULATORY

We aim to validate a toolset of novel incentive mechanisms for all stakeholders across the service delivery chain and layers, to induce the desired end-to-end coordinated energy aware efficient behaviour. While the broader vision includes all layers, our current focus is on

end-users. More precisely, concerning ICT service consumers we will develop mechanisms to motivate the “wise” consumption of services. Such mechanisms are based on **behavioural incentives** (i.e., social peer pressure, e.g., eco badges), potentially combined with economic incentives ([see Sections 2.1.3.6. and 2.1.4.6. of D1.4 \(ref\)](#)).

Incentive Mechanisms for Media streaming carbon footprint transparency

Here, we view UC3 and UC4 as interconnected. Thus, we are building a theoretical model and simulations for both UCs, covering both the behavioural (UC4) and economic (UC3) incentives. In fact, the use of behavioural incentives for the user can mitigate the need for economic ones. Also, we include - at least partly - UC7: carbon certificates in the modelling of UC3.

UC3 and UC4 are integrated, as behavioural incentives directly influence economic incentives—for instance, altruistic users require lower financial motivation compared to self-interested users. Similarly, for service providers, behavioural incentives can translate into economic advantages, such as enhancing their green image. In particular, the user-centric case presented here is common for both UC3 and UC4. It focuses on End-users, specifically ICT service consumers, and on their interaction with the OTT. Incentives are mainly translated into Environmental Points, which is used as a unified “currency”, that is a unified representation of the value of user's sustainable behaviour in both UCs. Incentives of other providers, although possibly affected by the selections of the user, are not addressed at this point.

Validation is both theoretical and simulation-based. We employ models for the various stakeholders and the objectives and behaviour of both the service providers and the end user. The validation metrics/KPIs include (depending on the case), a selection from the following: the carbon footprint reduction, the reduction of energy consumption, the cost for implementing the incentive mechanisms (including cost for collecting necessary measurements), and the sensitivity/robustness of the above performance metrics to different system parameters and stakeholder behaviours.

Incentive Mechanisms for Carbon Aware AI¶

Next we move to **Incentive Mechanism for Carbon Aware AI**, that is UC10 in **EXIGENCE project**. In Deliverables D1.1, D1.2 and D3.1, the initial design of UC10 outlined an incentive framework targeting multiple stakeholders, e.g., end-users, AI service providers, energy suppliers, and network service providers. Incentives ranged from behavioral rewards and discounts for individuals and firms to operational savings and compliance advantages for providers. These mechanisms remained descriptive and conceptual.¶

Here we present a model that advances the AI-Training framework by providing a formal game-theoretic representation of the interaction between end-users and the AI service provider. The model assumes a federated learning setting where a central AI-service provider coordinates training with multiple end-users, each choosing the number of data samples to contribute and whether to participate, thereby creating a collective demand for energy. The total available renewable (green) energy is limited and shared among end-users, while any excess demand is met by the grid at higher cost. Provider's coordination relies on incentivization, that is rewards and penalties, with respect to the system's green energy constraints. The framework

assumes rational participants, that act simultaneously. The outcome of this interaction reflects stable allocations of data and efficient energy use.

Moving to AI-Inference, the proposed model explores how AI services can lower their carbon footprint by slightly adjusting the accuracy or speed of their responses, and how users with different preferences react to such changes. It highlights that people value AI services in different ways--some care most about precision, some about rapid answers, and others about making environmentally responsible choices. By understanding these differences, the study examines how to encourage users to accept greener service options in ways that still feel fair and reasonable to them. It also considers how the environmental impact of AI varies across locations and times, and how this can influence when and how greener settings are offered.

Preliminary experiments demonstrate several critical dynamics. Without penalties, en-users may abstain or resort to non-green energy. High penalties shift behavior toward renewable use but risk discouraging participation, affecting the overall performance. End-users with disproportionately large data volumes dominate training, raising fairness concerns. Average play across repeated rounds tends to stabilize around allocations that maximize the use of available green energy. Moving to AI-Inference, the analysis finds that the proposed incentives can be introduced in flexible ways, such as offering periods of the day or small portions of a user's requests at slightly lower quality, depending on energy conditions. Users who are more environmentally conscious tend to respond more positively to such changes, while others may require stronger incentives. The results suggest that AI providers can reduce emissions meaningfully by tailoring service tiers, aligning them with user preferences, and taking advantage of differences in carbon intensity across data centers or time periods. This approach allows for sustainability improvements without heavily compromising the user experience.

~~In Deliverables D1.1, D1.2 and D3.1, the initial design of UC10 outlined an incentive framework targeting multiple stakeholders, e.g., end-users, AI service providers, energy suppliers, and network service providers. Incentives ranged from behavioural rewards and discounts for individuals and firms to operational savings and compliance advantages for providers. These mechanisms remained descriptive and conceptual.~~

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~~End-users' actions translate into energy demand, with the allocation of green energy versus grid energy explicitly modeled. Therefore, each end-user's utility depends on three components: (i) *accuracy* of the global model, (ii) *proximity* to available renewable energy, and (iii) *quality of service*. Costs are structured to reflect both linear and quadratic terms in green energy use, penalizing over-consumption relative to peers. The AI service provider's strategy~~

is to set a penalty parameter (γ) that balances participation incentives with the system's green energy constraints. The AI service provider's utility integrates model accuracy, participation rates, and efficient use of the renewable energy budget. The result is a normal-form game in which Nash equilibria (or their approximations) determine the stable allocation of samples, energy use, and penalties.¶¶

Additionally, the model benefits end-users with better performance to experience a higher level of QoS and receive a larger reward. At the same time, it integrates financial costs for the energy consumption, utilizing well-defined mathematical formulas [18], with green energy being much cheaper than grid energy.¶¶

Formally, the incentives for end-users are captured by the following formula¶¶

$$\mathbb{¶}$$

For each end-user i , ω_i is a binary variable indicating its participation. If $\omega_i = 1$, the end-user participates, receives a profit P_i and incurs a cost C_i . Otherwise, end-user i experiences a penalty γ .¶¶

On the other hand, the incentives for the AI service provider are defined by the equation¶¶

$$\mathbb{¶}$$

The utility that the AI provider experiences depends positively on the performance of each participating user, expressed as $\omega_i \Phi_i$, and negatively on both i) the number of non-participations and ii) the unused portion of green energy, captured by the function $F(\omega, \gamma)$. The latter is corroborated by the negative impact of unused green energy in the system [19].¶¶

In our model, there are two major incentives. The first is the penalty γ that is imposed by the AI service provider on end-users who either abstain from the training or are careless with green energy. Through this mechanism, the provider seeks to guide the system towards higher participation and lower non-green energy consumption. However, there is a trade-off: a small penalty may lead some end-users to avoid participation or to overuse grid energy, slowing down model's convergence, or undermining sustainability. Conversely, a large penalty may push the end-users to contribute only a small amount of data, thereby limiting training performance.¶¶

The second incentive arises from energy costs. If an end-user consumes only green energy, its cost remains low, while the use of grid energy results in much higher expenses. This encourages end-users to minimize non-green energy usage. Furthermore, if a single end-user consumes a disproportionately large share of the available green energy, it faces a higher cost. This discourages oligopolistic behaviour and improves fairness in the distribution of green resources.¶¶

Moreover, our model has been validated with both synthetic data and real-world datasets. The use of synthetic data enables us to systematically explore the behavior of the model under controlled conditions and stress-test the system across a wide range of scenarios. The inclusion of real-world datasets provides evidence of practical applicability, ensuring that the model's predictions remain meaningful in realistic environments where data heterogeneity, participation incentives, and renewable energy availability are more complex. Notably, in

~~several experimental settings, we observe that the proposed mechanism leads the interaction to almost zero carbon emission outcome.~~^[1]

Preliminary experiments demonstrate several critical dynamics. Without penalties, end-users may abstain or resort to non-green energy. High penalties shift behaviour toward renewable use but risk discouraging participation, affecting the overall performance. End-users with disproportionately large data volumes dominate training, raising fairness concerns. Average play across repeated rounds tends to stabilize around allocations that maximize the use of available green energy. The model also captures the strategic role of the AI service provider in adjusting γ to optimize both efficiency and participation.

~~Concerning AI inference services in UC10, we have investigated the tradeoffs between inference accuracy and response latency and developed user incentives for reducing the associated carbon emissions while taking these tradeoffs into account. We have introduced a novel user model that captures preferences for QoE (Quality of Experience) in two dimensions: inference accuracy and inference response latency, while also accounting for a user's degree of environmental consciousness. Specifically, the user's valuation for inference accuracy is modeled as an increasing concave function of the carbon emissions and the user's valuation for inference response latency is modeled as a horizontally flipped sigmoid function of the carbon emissions. The relative importance of inference accuracy compared to latency is reflected by a weight parameter, which can reflect whether the user is accuracy-sensitive or latency-sensitive. Finally, the level of a user's environmental consciousness is modeled through different utility functions, whereby a user with high environmental consciousness achieves a higher utility for the same inference accuracy and for the same response latency, compared to a user with lower environmental consciousness.~~^[1]

~~In the proposed framework, the user model described above is combined with a model of an AI inference service provider that captures the carbon emissions reduction achieved when the provider offers lower inference accuracy and higher inference response latency. An AI inference service provider can offer different inference accuracies by utilizing multiple trained AI models with a different number of parameters and layers, which have a different energy consumption and corresponding carbon emissions. The tradeoff between inference accuracy and carbon emissions can be different for different application domains. Additionally, an AI inference provider can offer multiple inference response latencies by sharing GPU resources among multiple inference requests. More resource sharing will result in higher inference latency but will provide the benefit of reduced carbon emissions.~~^[1]

~~Our framework can be used to investigate the following questions:~~^[1]

- ~~• What combination of AI inference accuracy reduction and AI inference latency increase is best to achieve a target carbon emission reduction? "Best" here refers to the combination that provides the target carbon emissions reduction with the smallest decrease of the user's utility. The answer depends both on a user's individual valuation for inference accuracy and for inference response latency and the relative evaluation of these two QoE metrics, i.e. whether the user is accuracy-conscious or latency-conscious.~~^[1]
- ~~• What incentives need to be provided to achieve a target carbon emissions reduction? The answer to this question needs to consider the behaviour of a rational user that~~

seeks to maximize his net benefit together with service provider QoE model that captures the tradeoff of inference accuracy and response latency on one hand, and the carbon emissions on the other. Indeed, the best combination of inference accuracy reduction and response latency increase to achieve a specific carbon emissions reduction depends on the user's relative evaluation for the two QoE parameters.¶¶

- How do the above depend on user's environmental consciousness. Such as a high-quality versus a green user?¶¶

Our analysis provides relative user incentives that achieve a specific outcome. The actual incentives will use a scaling factor to convert these relative values to monetary units (or environmental points).¶¶

Our framework together with the carbon intensity in different geographic locations and in different time periods allows us to understand how the carbon intensity influences incentives. As expected, when within one geographic area or during a particular time period the carbon intensity is low, i.e. the amount of carbon emissions for a specific amount of energy consumption is low due to the use of renewable energy sources, then the incentives can be low. Moreover, the level of incentives also depends on the user's valuation for inference accuracy and latency, as well as on the user's environmental consciousness. Hence, the same level of incentives can be more effective, i.e. result in a higher reduction of carbon emissions for a particular time period with high carbon intensity, for an environmentally conscious user compared to a user who seeks the highest inference accuracy and the lowest latency. If an AI service provider can respond to queries from data centers located in different locations and having different (uncorrelated) carbon intensities, we show how the provider can exploit the capability of directing AI queries to different locations to minimize the incentives it needs to provide to its users so that they reduce their requests for high accuracy and low latency in order to reduce the corresponding carbon emissions.¶¶

By combining the analysis of historic carbon intensity data with our model, we can propose different service tiers that incorporate different tradeoffs in inference accuracy and latency that are more appropriate for particular types of users. In this direction, we propose two types of tiered contracts:¶¶

- Time zone based two-tier contract: With such a contract, inference responses during specific time periods in a day may (but not necessarily) be provided in a lower tier, which corresponds to a lower inference accuracy and a higher response latency. The decision to respond to queries in the lower quality tier, within the specific time periods of the day, is taken by the provider and can depend on the actual carbon intensity (whereas the specific time zones are a priori determined based on the forecasted carbon intensity that considers historic data) and the actual AI inference demand.¶¶
- Two-tier contract with fixed tier percentages: Within some period, such as a day or a month, up to some percentage of inference responses can be provided in a lower tier, with a lower inference accuracy and a higher inference response latency. For example, up to 20% of the inference query can be provided (but are not necessarily provided) at a lower quality tier within the period of a day or a month. As above, the decision to respond to queries in the lower quality tier is taken by the provider, based on the actual carbon intensity and the AI inference demand, but must be below the maximum percentage threshold.¶¶

Finally, our framework can be extended in the case of users that have a varying valuation of inference accuracy and latency during their query session, e.g. a user can start with initially generic queries requiring low accuracy and accepting high latency, while for later queries the user requires higher accuracy and is less tolerant to delay.¶¶

Table 5 Computational perspectives of Incentive Mechanisms Across 6G Value Network Domains¶¶

UC	Approach	Complexity	Input	Computational time
UC3-4	User-Acceptance Model	Linear	No strategic agents	~0.33s
	Gamified Incentives	Complexity of Stack. eq.	OTT's strategic space	~1s
	AI-training	Complexity of Nash eq.	OTT's and users' strategic space	Intractable
UC10	AI-training	Complexity of Heuristic	OTT's and users' strategic space	~22s*

In Table 5, we present several computational considerations related to the methods we used to construct the incentive mechanisms in UC 3, 4, and 10. As benchmark input, we consider a population of 1000 users and one OTT. In the first case, the user acceptance model for video streaming, there are no strategic agents, since the users simply react to the incentives offered by the OTT (i.e., the provider). For the same scenario, the problem of gamified incentivization is as hard as computing a Stackelberg equilibrium. Here, both the agents and the OTT are strategic, meaning that they adapt their decisions in response to the decisions of the other participants. More specifically, each user chooses between high and low bitrate video resolutions, while the OTT sets the offered incentives and the parameters of the Stackelberg game; that is, the serious game.¶¶

Next, in the case of AI training, the users are incentivized through a finite normal form game. Users choose whether to participate and the training intensity they will employ during local training. The OTT (i.e., the server) selects the rewards/penalties policy. The natural solution concept here is a Nash equilibrium; therefore, the complexity of the incentive mechanism corresponds to the complexity of computing a Nash equilibrium in a finite normal form game, which is intractable for an arbitrary number of players. To mitigate this issue, we construct a heuristic that users employ to make decisions. In particular, each user runs a fictitious play algorithm for 1000 iterations, initialized from the same starting point for each user, and then acts according to the resulting strategy. Thus, the complexity of the incentive mechanisms reduce to the complexity of the heuristic. In the bottom cell of the right column, we report that the runtime of mechanism incentives with heuristics for UC10 is approximately 22s; however, this evaluation was conducted with a smaller number of users due to memory limitations. To address this, we will rely on parallelization. ¶¶

¶¶

Even though Nash equilibrium is an appealing and universal solution concept in multi-agent strategic interactions, it comes with several issues. One is computational hardness, which we mitigate through heuristics that compute approximate solutions. A second issue is the potential multiplicity of Nash equilibria. Since this is inherent to multi-agent strategic interactions, we inherit it as well. However, if all participants converge to an approximate Nash equilibrium, then approximately neither will deviate from this choice. Thus, multiplicity cares us about the quality of the outcome. We can treat this by examining the different outcomes of the heuristic when it is utilized at various points in the strategic space.¶

4.2 UPDATES ON USE CASE-SPECIFIC DESIGNS

We summarize the current adaptations and planned extensions of the incentive mechanisms across the different use cases (UC3, UC4, UC10, etc.). At this stage the most substantive updates concern UC3 and UC4, where the incentive framework has been refined in greater detail. For UC10 we provide the complete incentive model in Section 4.1, while the full analysis and experimental evaluation are planned for inclusion in upcoming project deliverables.

Incentive Mechanisms for Media streaming carbon footprint transparency

We begin by presenting the functional architecture for the user-centric scenario (Figure 6) together with an accompanying flowchart (Figure 7). Building on this foundation, we then outline evaluation methods designed to capture and analyze users' responsiveness to incentives and their patterns of acceptance.

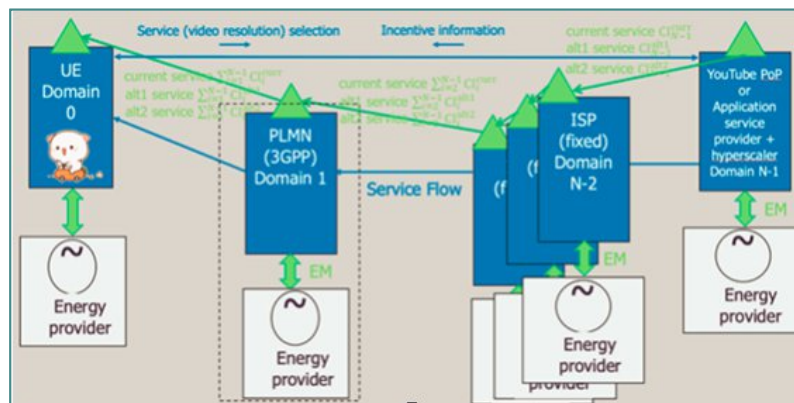


Figure 6 Functional Architecture Instantiated for user-centric case.

Figure 6Figure 6 Functional Architecture Instantiated for user-centric case. shows how user-centric case maps the EXIGENCE Functional Architecture. This instantiation involves i) the provision of a service (e.g., video streaming) from a service provider (e.g., YouTube), via one or more fixed Internet Service Providers (ISPs) and a mobile network operator (PLMN) to an end-user – User Equipment (UE), ii) energy monitoring from an energy provider, and iii) incentive information again from the Service Provider to the UE.

The architecture spans several domains: starting with the UE which communicates through a rightward-flowing service path across the network, with the application service provider. In parallel, a leftward flow of incentive information provides the UE with various streaming alternatives, each annotated with an aggregated energy or carbon cost, helping UEs to make

informed choices. Furthermore, each domain in the architecture is linked to its respective energy provider, which represents real-time tracking of energy consumption.

Observe that the content delivery follows along the service arrows (blue arrows in the figure) from the provider to the UE. The carbon footprint information (green arrows in the figure) also flows from the provider to the UE. The UE's connection with the energy provider could be also considered in this case, as the energy efficiency related to video streaming can also result in device energy efficiency. The overall objective is to enhance environmentally conscious behaviour. For that, we introduce and utilize exchange of energy and carbon-related information across all layers. The architecture enables stakeholders to monitor and optimize the carbon footprint of streaming sessions, thus heading to more sustainable service delivery across chain.

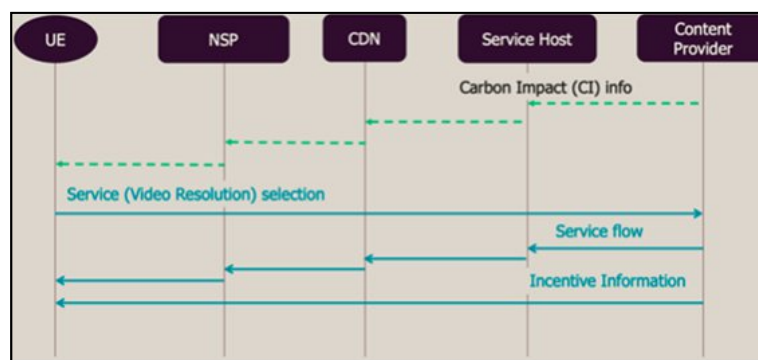


Figure 7 Flowchart for user-centric case.

The flowchart for user-centric case illustrates the flow of carbon-aware incentive-based information in digital environments. Specifically, the Content Provider sends Carbon Impact (CI) information to the Service Host, which is then propagated back to the UE via the Content Delivery Network (CDN) and the Network Service Provider (NSP). The UE selects the service (e.g., video resolution) based on this information, triggering a service flow back through the chain. In parallel, the UE has access to Incentive Information as provided by the Content Provider. The whole setup encourages users to choose lower-energy and consequently lower carbon footprint options, both by enabling incentives and by making environmental impacts visible.

Proposed Evaluation Methods for building the model for user's "acceptance" and "responsiveness" of incentives.

We present the proposed (so far) methods for modelling the evaluation. The actual modelling and the simulation results will be presented in D3.2. The proposed user acceptance model focuses on assessing the incentivization of energy-efficient behaviour in video streaming, by modelling the extent of acceptance by the users of the incentives offered. In particular, by means of incentives in place, users are encouraged to opt for lower video bitrates/resolutions through environmental points, which are converted into behavioural and/or economic rewards [20] [21]. By integrating gamification elements, such as recognizing monthly "champions" who save the most energy, the incentives scheme aims to increase user motivation. According to the user acceptance model, the decision-making process amounts to a binary choice framework; namely, users either accept the incentives and reduce energy consumption or maintain their usual viewing habits. This approach establishes a direct link between incentives

and energy reduction leading to lower carbon emissions and flexibility in the grid. ~~For instance, in case where the incentivizations shifts the users choices from Ultra HD to Full HD, the application of the user acceptance model may lead to up to 1515kWh.~~

To evaluate the trade-off between user experience and environmental impact, the model includes a “greenness factor”, which works as follows. The Quality of Experience (QoE) assessment quantifies the utility loss associated with reduced video bitrate using Mean Opinion Score (MOS). By introducing the concept of the “greenness factor,” the model accounts for user willingness to prioritize energy efficiency. Additionally, uncertainty in user acceptance of the incentives is incorporated by means of a probabilistic approach, thus modelling the binary decision as a Bernoulli trial.

Further considerations include altruistic behaviour, where users balance personal satisfaction with social wellbeing. Altruistic users require smaller incentives to adopt eco-friendly practices, as they derive additional utility from contributing to sustainability. Over time, behavioural education can shift user preferences, reducing reliance on external rewards. In our validation approach, we employ the Social Green-Wellbeing (SWy) metric, which aggregates the utilities of environmentally conscious users.

The greenness factor and altruistic behaviour with SWy are actually interconnected, since they both influence a user's willingness to adopt energy-efficient streaming options, but from different perspectives—individual choice and collective impact. The greenness factor represents a user's personal valuation of energy efficiency in their QoE assessment. Users with a high greenness factor are more likely to accept energy-efficient options even with minimal incentives. Their decision is driven by an intrinsic motivation towards sustainability rather than external rewards. On the other hand, altruistic behaviour, as modelled with Social Green-Wellbeing (SWy), captures how eco-conscious users collectively contribute to a broader social benefit. Altruistic users weigh personal satisfaction against societal well-being, meaning they require lower incentives to make energy-efficient choices. As more users adopt altruistic behaviours, the aggregated SWy metric enables us to quantify and optimize incentive structures, ensuring that the environmental impact reduction aligns with user motivations. Together, the greenness factor and SWy create a reinforcing loop: users with higher greenness factors naturally contribute to SWy, while behavioural education and incentives enhance users' altruistic tendencies, shifting their preferences toward more sustainable actions with lower economic motivation.

The steps depicted in Figure 8 are connected in a logical flow to create an effective user acceptance model for energy-efficient video streaming. The entire framework works in a cycle, where incentives and education gradually shape user choices, optimizing sustainability while maintaining satisfaction. The proposed user acceptance model is implemented as a modular software component within the end-to-end architecture. The model can interoperate with existing QoE monitoring tools and energy measurement data sources, accounting for both individual user preferences and social impact. Importantly, incentives and feedback are delivered to the client, agglomerating real-time service usage parameters (e.g., video resolution, device type) and historical data. Gamification elements such as leaderboards can be depicted via a UI component. The incentives component must be orchestration-agnostic, arranged for scalability, and will be able to obtain energy consumption and CO₂e data from

EXIGENCE measurement components and other data (e.g., MOS estimation, carbon intensity) from external sources.

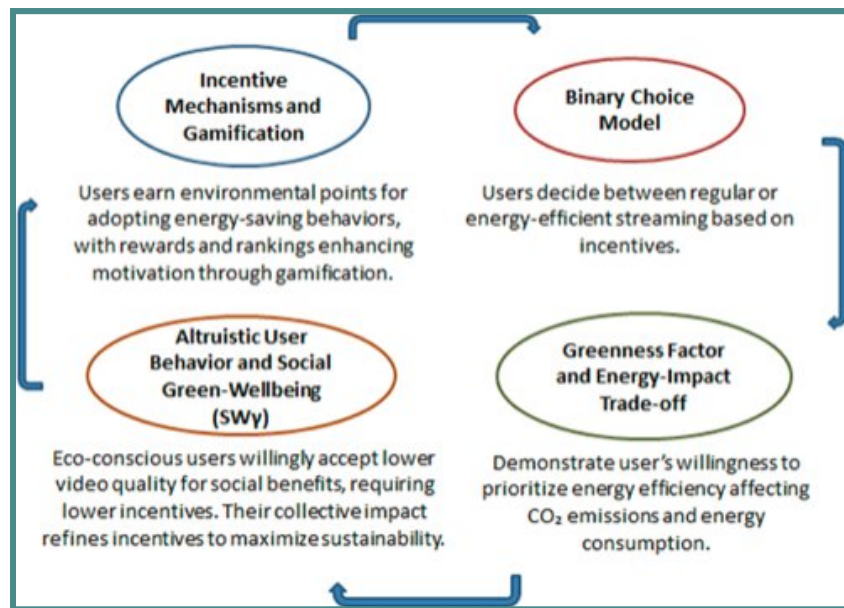


Figure 8 Effective user acceptance model for energy-efficient video streaming.¶¶

¶¶

Incentive Mechanisms for Carbon-Aware AI¶¶

In Deliverables D1.1, D1.2 and D3.1, the initial design of UC10 outlined an incentive framework targeting multiple stakeholders, e.g., end-users, AI service providers, energy suppliers, and network service providers. Incentives ranged from behavioural rewards and discounts for individuals and firms to operational savings and compliance advantages for providers. These mechanisms remained descriptive and conceptual.¶¶

Here we present a model that advances the AI training framework by providing a formal game-theoretic representation of the interaction between end-users and the AI service provider. The model assumes a federated learning setting where a central AI-service provider coordinates training with multiple end-users, each choosing the number of data samples to contribute and whether to participate, thereby creating a collective demand for energy. The total available renewable (green) energy is limited and shared among end-users, while any excess demand is met by the grid at higher cost. The framework assumes rational participants that act simultaneously. The outcome of this interaction reflects stable allocations of data and efficient energy use.¶¶

End-users' actions translate into energy demand, with the allocation of green energy versus grid energy explicitly modeled. Therefore, each end-user's utility depends on three components: (i) *accuracy* of the global model, (ii) *proximity* to available renewable energy, and (iii) *quality of service*. Costs are structured to reflect both linear and quadratic terms in green energy use, penalizing over-consumption relative to peers. The AI service provider's strategy is to set a penalty parameter (γ) that balances participation incentives with the system's green energy constraints. The AI service provider's utility integrates model accuracy, participation rates, and efficient use of the renewable energy budget. The result is a normal-form game in

which Nash equilibria (or their approximations) determine the stable allocation of samples, energy use, and penalties.

Additionally, the model benefits end-users with better performance to experience a higher level of QoS and receive a larger reward. At the same time, it integrates financial costs for the energy consumption, utilizing well-defined mathematical formulas [18], with green energy being much cheaper than grid energy.

Formally, the incentives for end-users are captured by the following formula

$$\omega_i(P_i - C_i) - (1 - \omega_i)\gamma$$

For each end-user i , ω_i is a binary variable indicating its participation. If $\omega_i = 1$, the end-user participates, receives a profit P_i and incurs a cost C_i . Otherwise, end-user i experiences a penalty γ .

On the other hand, the incentives for the AI service provider are defined by the equation

$$\sum_{i=1}^n (\omega_i \Phi_i - (1 - \omega_i)\gamma) - \Gamma(\omega, \gamma)$$

The utility that the AI provider experiences depends positively on the performance of each participating user, expressed as $\omega_i \Phi_i$, and negatively on both i) the number of non-participations and ii) the unused portion of green energy, captured by the function $\Gamma(\omega, \gamma)$. The latter is corroborated by the negative impact of unused green energy in the system [19].

In our model, there are two major incentives. The first is the penalty γ that is imposed by the AI service provider on end-users who either abstain from the training or are careless with green energy. Through this mechanism, the provider seeks to guide the system towards higher participation and lower non-green energy consumption. However, there is a trade-off: a small penalty may lead some end-users to avoid participation or to overuse grid energy, slowing down model's convergence, or undermining sustainability. Conversely, a large penalty may push the end-users to contribute only a small amount of data, thereby limiting training performance.

The second incentive arises from energy costs. If an end-user consumes only green energy, its cost remains low, while the use of grid energy results in much higher expenses. This encourages end-users to minimize non-green energy usage. Furthermore, if a single end-user consumes a disproportionately large share of the available green energy, it faces a higher cost. This discourages oligopolistic behaviour and improves fairness in the distribution of green resources.

Moreover, our model has been validated with both synthetic data and real-world datasets. The use of synthetic data enables us to systematically explore the behavior of the model under controlled conditions and stress-test the system across a wide range of scenarios. The inclusion of real-world datasets provides evidence of practical applicability, ensuring that the model's predictions remain meaningful in realistic environments where data heterogeneity, participation incentives, and renewable energy availability are more complex. Notably, in several experimental settings, we observe that the proposed mechanism leads the interaction to almost zero carbon-emission outcome.

Preliminary experiments demonstrate several critical dynamics. Without penalties, end-users may abstain or resort to non-green energy. High penalties shift behaviour toward renewable use but risk discouraging participation, affecting the overall performance. End-users with disproportionately large data volumes dominate training, raising fairness concerns. Average play across repeated rounds tends to stabilize around allocations that maximize the use of available green energy. The model also captures the strategic role of the AI service provider in adjusting γ to optimize both efficiency and participation.¶

Concerning AI inference services in UC10, we have investigated the tradeoffs between inference accuracy and response latency and developed user incentives for reducing the associated carbon emissions while taking these tradeoffs into account. We have introduced a novel user model that captures preferences for QoE (Quality of Experience) in two dimensions: inference accuracy and inference response latency, while also accounting for a user's degree of environmental consciousness. Specifically, the user's valuation for inference accuracy is modeled as an increasing concave function of the carbon emissions and the user's valuation for inference response latency is modeled as a horizontally flipped sigmoid function of the carbon emissions. The relative importance of inference accuracy compared to latency is reflected by a weight parameter, which can reflect whether the user is accuracy-sensitive or latency-sensitive. Finally, the level of a user's environmental consciousness is modeled through different utility functions, whereby a user with high environmental consciousness achieves a higher utility for the same inference accuracy and for the same response latency, compared to a user with lower environmental consciousness.¶

In the proposed framework, the user model described above is combined with a model of an AI inference service provider that captures the carbon emissions reduction achieved when the provider offers lower inference accuracy and higher inference response latency. An AI inference service provider can offer different inference accuracies by utilizing multiple trained AI models with a different number of parameters and layers, which have a different energy consumption and corresponding carbon emissions. The tradeoff between inference accuracy and carbon emissions can be different for different application domains. Additionally, an AI inference provider can offer multiple inference response latencies by sharing GPU resources among multiple inference requests. More resource sharing will result in higher inference latency but will provide the benefit of reduced carbon emissions.¶

Our framework can be used to investigate the following questions:¶

- What combination of AI inference accuracy reduction and AI inference latency increase is best to achieve a target carbon emission reduction? “Best” here refers to the combination that provides the target carbon emissions reduction with the smallest decrease of the user's utility. The answer depends both on a user's individual valuation for inference accuracy and for inference response latency and the relative evaluation of these two QoE metrics, i.e. whether the user is accuracy conscious or latency conscious.¶
- What incentives need to be provided to achieve a target carbon emissions reduction? The answer to this question needs to consider the behaviour of a rational user that seeks to maximize his net benefit together with service provider QoE model that captures the tradeoff of inference accuracy and response latency on one hand, and the carbon emissions on the other. Indeed, the best combination of inference accuracy

reduction and response latency increase to achieve a specific carbon emissions reduction depends on the user's relative evaluation for the two QoE parameters.¶

- How do the above depend on user's environmental consciousness. Such as a high-quality versus a green user?¶

Our analysis provides relative user incentives that achieve a specific outcome. The actual incentives will use a scaling factor to convert these relative values to monetary units (or environmental points).¶

Our framework together with the carbon intensity in different geographic locations and in different time periods allows us to understand how the carbon intensity influences incentives. As expected, when within one geographic area or during a particular time period the carbon intensity is low, i.e. the amount of carbon emissions for a specific amount of energy consumption is low due to the use of renewable energy sources, then the incentives can be low. Moreover, the level of incentives also depends on the user's valuation for inference accuracy and latency, as well as on the user's environmental consciousness. Hence, the same level of incentives can be more effective, i.e. result in a higher reduction of carbon emissions for a particular time period with high carbon intensity, for an environmentally conscious user compared to a user who seeks the highest inference accuracy and the lowest latency. If an AI service provider can respond to queries from data centers located in different locations and having different (uncorrelated) carbon intensities, we show how the provider can exploit the capability of directing AI queries to different locations to minimize the incentives it needs to provide to its users so that they reduce their requests for high accuracy and low latency in order to reduce the corresponding carbon emissions.¶

By combining the analysis of historic carbon intensity data with our model, we can propose different service tiers that incorporate different tradeoffs in inference accuracy and latency that are more appropriate for particular types of users. In this direction, we propose two types of tiered contracts:¶

- Time-zone based two-tier contract: With such a contract, inference responses during specific time periods in a day may (but not necessarily) be provided in a lower tier, which corresponds to a lower inference accuracy and a higher response latency. The decision to respond to queries in the lower quality tier, within the specific time periods of the day, is taken by the provider and can depend on the actual carbon intensity (whereas the specific time zones are a priori determined based on the forecasted carbon intensity that considers historic data) and the actual AI inference demand.¶
- Two-tier contract with fixed tier percentages: Within some period, such as a day or a month, up to some percentage of inference responses can be provided in a lower tier, with a lower inference accuracy and a higher inference response latency. For example, up to 20% of the inference query can be provided (but are not necessarily provided) at a lower quality tier within the period of a day or a month. As above, the decision to respond to queries in the lower quality tier is taken by the provider, based on the actual carbon intensity and the AI inference demand, but must be below the maximum percentage threshold.¶

Finally, our framework can be extended in the case of users that have a varying valuation of inference accuracy and latency during their query session, e.g. a user can start with initially

generic queries requiring low accuracy and accepting high latency, while for later queries the user requires higher accuracy and is less tolerant to delay.

Computational Perspectives of Incentive Mechanisms Incentive Mechanisms for Carbon Aware AI

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Table 5 Computational perspectives of Incentive Mechanisms Across 6G Value Network Domains

UC	Approach	Complexity	Input	Computational time
UC3-4	User-Acceptance Model	Linear	No strategic agents	~0.33s
	Gamified Incentives	Complexity of Stack. eq.	OTT's strategic space	~1s
	AI-training	Complexity of Nash eq.	OTT's and users' strategic space	Intractable
UC10	AI-training	Complexity of Heuristic	OTT's and users' strategic space	~22s*

¶

In Table 5, we present several computational considerations related to the methods we used to construct the incentive mechanisms in UC 3, 4, and 10. As benchmark input, we consider a population of 1000 users and one OTT. In the Media Streaming first case, considering the user acceptance model for video streaming, there are no strategic agents, since the users simply react to the incentives offered by the OTT (i.e., the provider). As benchmark input, we consider a population of 1000 users and one OTT. For the same scenario, the problem of gamified incentivization is as hard as computing a Stackelberg equilibrium. Here, both the agents and the OTT are strategic, meaning that they adapt their decisions in response to the decisions of the other participants. More specifically, each user chooses between high and low bitrate video resolutions, while the OTT sets the offered incentives and the parameters of the Stackelberg game; that is, the serious game. To evaluate our model in real-world situations, we consider the case where the N users have to choose between 20Mbps bitrates, that is UltraHD video streaming resolution, and 5Mbps bitrates, that is FullHD video streaming resolution. The proposed user acceptance model achieved total energy reduction almost 67%.

¶

Next, in the case of AI-training, the users are incentivized through a finite normal-form game. Users choose whether to participate and the training intensity they will employ during local training. The OTT (i.e., the server) selects the rewards/penalties policy. The natural solution concept here is a Nash equilibrium; therefore, the complexity of the incentive mechanism corresponds to the complexity of computing a Nash equilibrium in a finite normal-form game, which is intractable for an arbitrary number of players. To mitigate this issue, we construct a heuristic that users employ to make decisions. In particular, each user runs a fictitious-play algorithm for 1000 iterations, initialized from the same starting point for each user, and then acts according to the resulting strategy. Thus, the complexity of the incentive mechanisms

reduces to the complexity of the heuristic. In the bottom cell of the right column, we report that the runtime of mechanism incentives with heuristics for UC10 is approximately 22s; however, this evaluation was conducted with a smaller number of users due to memory limitations. To address this, we will rely on parallelization. ¶

4.2.1 CARBON CREDIT AS INCENTIVE MECHANISM

Economic instruments have emerged as the effective and cost-efficient tools for achieving emissions reductions globally, offering a shift from traditional command-and-control methods to market-based approaches that leverage financial incentives to encourage behavioural change. Central to economical instruments are carbon pricing mechanisms, particularly emissions trading systems and carbon taxes, form the backbone of global climate policy, while innovative mechanisms like biodiversity credits and nature performance bonds represent the cutting edge of environmental finance. Figure 9 shows the coverage provided by selected ETFs worldwide.

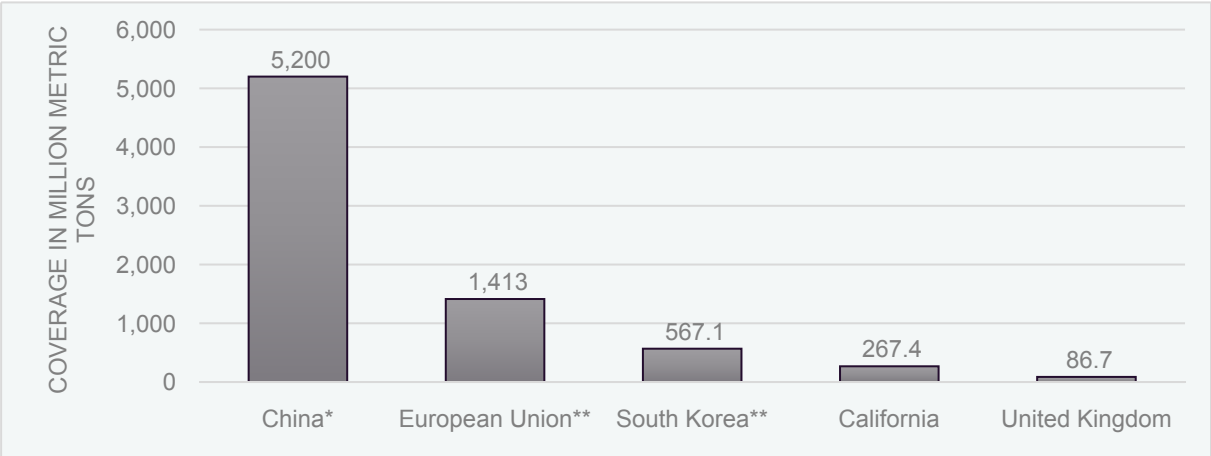


Figure 9: Emissions covered by selected emissions trading systems (ETS) worldwide as of 2025 (in million metric tons of carbon dioxide equivalent)

Carbon credits, each representing one metric ton of verified carbon dioxide equivalent (tCO₂e) emission reduced or removed, are traded as financial instruments in voluntary and compliance markets. Compliance credits are generated under mandatory cap-and-trade systems like the EU Emissions Trading System (EU ETS), which covers around 45% of the EU’s greenhouse gas emissions. Between 2005 and 2023, the EU ETS achieved a 47% emissions reduction, highlighting the effectiveness of regulatory carbon markets.

Compliance Market Credits are generated under mandatory cap-and-trade systems like the European Union Emissions Trading System (EU ETS), which covers approximately **45% of the EU’s greenhouse gas emissions** Error! Reference source not found. **Error! Reference source not found..** Between 2005 and 2023, the EU ETS achieved a **47% reduction in emission**, demonstrating the effectiveness of regulatory carbon markets **Error! Reference source not found..** Figure 10 represents the forecasted pricing on the EU-ETS and phasing in of Carbon Border Adjustment Mechanism (CBAM), putting regulatory strain and financial impacts on businesses. The CBAM is a key EU climate policy tool designed to prevent **carbon**

leakage, i.e. the relocation of carbon-intensive production to countries with weaker climate regulations, to ensure fair carbon pricing for imported goods.

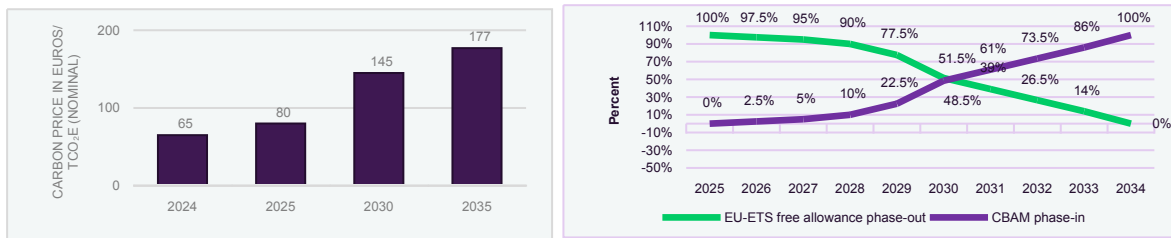


Figure 10: Forecast European Union Emissions Trading System (EU-ETS) average carbon allowance prices from 2024 to 2035 (in euros per metric ton of CO₂ equivalent) and Pathway of EU ETS free allowances phase-out and Carbon Border Adjustment Mechanism (CBAM) phase-in from 2025 to 2034 (in percent)

Carbon credits market prices are key incentivization mechanism in exigence, as outlined in deliverables D1.2 and D3.1. To implement Exigence's carbon credit-based incentive mechanisms, daily carbon market data must be retrieved via standardized API from a trustable external source. For instance, "Trading Economics" **Error! Reference source not found.** offers historical and (near) real-time EUA prices via a paid API subscription. **Error! Reference source not found.** illustrates the daily prices of EU Allowances (EUA), which are used later in our numerical evaluations. During the period from July 10, 2024, to July 8, 2025, UEUA prices ranged from €60.29 to €83.93 during the period.



Figure 11: EU Allowance market prices **Error! Reference source not found.**

The voluntary carbon market (VCM) reached approximately **\$2 billion in 2022** [1] with forestry and land-use projects alone accounting for **\$1.7 billion for more than 285 million tCO₂e** **Error! Reference source not found.**

VMC are created through independent crediting standards including the Verified Carbon Standard (Verra), Gold Standard, Climate Action Reserve, and American Carbon Registry. These credits enable companies and individuals to voluntarily offset their emissions beyond regulatory requirements **Error! Reference source not found.** **REDD+ Credits** represents the largest category in voluntary markets, **REDD+ credits account for approximately 40% of voluntary carbon market volume.** These credits finance forest conservation activities to

reduce emissions from deforestation and forest degradation in developing countries **Error! Reference source not found..**

Carbon credit prices vary significantly across markets and project types. Prices in the voluntary carbon market are rising but remain significantly lower than those in compliance markets, where EUA prices are regulated. The **average price of REDD+ credits increased from \$3.9/tCO₂ in 2019 to \$4.7/tCO₂ in 2021** **Error! Reference source not found.** In contrast, the EU ETS has occasionally experienced dramatic price increases, such as **exceeding €100 (\$108) per ton of CO₂ in February 2023, and as shown in** **Error! Reference source not found., reaching** €83.93 in February 2025.

The Intergovernmental Panel on Climate Change indicates that more robust mechanism in the carbon prices with consideration of **\$135–\$5,500 per ton CO₂ in 2030 and \$245–\$13,000 per ton in 2050** would be useful to limit warming to 1.5°C **Error! Reference source not found..**

Concerns have emerged over the quality of REDD+ credits, with studies indicating they may deliver only a fraction of their claimed climate benefits. Key challenges include additionality, permanence, leakage, and baseline accuracy.

Recent analysis has raised concerns about credit quality, particularly for REDD+ projects, with studies indicating they may deliver **only a small fraction of their claimed climate benefit** **Error! Reference source not found.****Error! Reference source not found..** Key challenges include **Additionality**: Ensuring projects wouldn't have occurred without carbon finance; **Permanence**: Guaranteeing long-term emission reductions; **Leakage**: Preventing displacement of emissions to other areas; **Baseline methodology**: Accurately measuring counterfactual scenarios. To ensure environmental integrity, robust Monitoring, Verification, and Reporting (MVR) processes are essential for validating project outcomes and maintaining trust in voluntary carbon markets.

Government Subsidies for Emissions Reduction

Scale and Scope

Government subsidies represent a **\$2.6 trillion annual expenditure globally** as of 2024, equivalent to **2.5% of global GDP** **Error! Reference source not found..** However, the distribution heavily favours fossil fuels, with **approximately 70% of energy-sector subsidies supporting fossil fuels** while only **20% support renewable power generation**.

Fossil Fuel Subsidy Reform

Reforming fossil fuel consumption subsidies in 32 countries could reduce global greenhouse gas emissions by 5.46 billion tonnes of CO₂ by 2030, equivalent to the annual emissions of about 1,000 coal-fired power plants **Error! Reference source not found..** This reform would also save governments nearly \$3 trillion cumulatively by 2030.

In some countries, subsidy reform could achieve emission reductions exceeding 35% of national emissions. Venezuela, Iraq, and Algeria could see reductions over 20% through fossil fuel subsidy reform alone.

Clean Energy Subsidies

Federal support for renewable energy in the United States more than doubled from \$7.4 billion in FY 2016 to \$15.6 billion in FY 2022. During FY 2016-2022, 46% of federal energy subsidies were associated with renewable energy.

Types of clean energy subsidies include:

- **Production Tax Credits:** Payments per unit of renewable energy generated
- **Investment Tax Credits:** Percentage of project costs provided as tax relief
- **Loan Guarantees:** Government backing for private financing
- **Direct Grants:** Upfront capital support for projects
- **Accelerated Depreciation:** Enhanced tax write-offs for clean energy investments

Subsidy Design and Effectiveness

Research demonstrates that government subsidies have significant incentive effects on carbon emissions reduction **Error! Reference source not found..** Key design principles include:

- **Results-based payments:** Linking subsidies to verified outcomes
- **Sunset clauses:** Automatic phase-out as technologies mature
- **Technology neutrality:** Supporting best-performing solutions
- **Additionality requirements:** Ensuring subsidies drive incremental action

Recent studies show that **emissions reduction (ER) subsidies can always bring higher social welfare compared to no government subsidy scenarios** [9].

Complete Global Inventory of Economic Instruments

Carbon Pricing Mechanisms

Emissions Trading Systems: as already mentioned, ETS is one of the cornerstones of global carbon policy, with 36 carbon trading systems operational worldwide covering 18% of global emissions **Error! Reference source not found..** Major systems include the EU ETS, California's cap-and-trade program, and China's national ETS.

Carbon Taxes: Implemented in over **30 jurisdictions globally**, providing price certainty and broad coverage across economic sectors **Error! Reference source not found..** Carbon taxes offer comparative simplicity in implementation while ensuring comprehensive emission coverage.

Hybrid Systems: Combining carbon taxes and trading mechanisms to capture benefits of both approaches, including **price certainty and flexibility** **Error! Reference source not found..**

Certificates and Credits Systems

Renewable Energy Certificates (RECs): RECs are tradable certificates that represent proof that 1 megawatt-hour (MWh) of electricity was generated from renewable sources **Error! Reference source not found..** They are operating in 29 U.S. states plus the District of Columbia and Puerto Rico under renewable portfolio standards.

White Certificates: Energy efficiency certificates operating in France, Italy, Poland, and Spain, with France's program generating cumulative energy savings of over 2,400 TWh by 2021 **Error! Reference source not found..**

Biodiversity Credits: An emerging market valued at \$8 million currently, with projections of \$2 billion by 2030 and \$69 billion by 2050 **Error! Reference source not found..**

Feed-in Mechanisms

Feed-in Tariffs: Implemented in **over 50 countries globally** as of 2019, providing guaranteed long-term contracts (15-25 years) for renewable energy producers **Error! Reference source not found..** The German Renewable Energy Act has been particularly successful in driving renewable energy deployment.

Net Metering: Enabling distributed renewable energy adoption by crediting excess generation fed back to the grid **Error! Reference source not found..**

Financial Instruments

Green Bonds: The market reached **\$670.9 billion in aligned annual volume in 2024**, with the World Bank having issued **over \$20 billion equivalent through more than 230 bonds in 28 currencies** since 2008 **Error! Reference source not found.Error! Reference source not found..**

Environmental Performance Bonds: Used in extractive industries and large-scale projects to guarantee funds for environmental restoration and cleanup **Error! Reference source not found.Error! Reference source not found..**

Payment Systems

Payment for Ecosystem Services (PES): Market-based approach providing economic incentives for ecosystem conservation, implemented across **watersheds, biodiversity conservation, carbon sequestration, and landscape beauty** **Error! Reference source not found.**

REDD+ Payments: Results-based payments for verified emission reductions from forest conservation, with potential annual value of **\$1.7 billion in voluntary carbon markets** **Error! Reference source not found..**

Regulatory Instruments

Deposit-Refund Systems: Implemented in **10 European countries** with impressive results - **Norway achieves 97% recycling rate for plastic bottles**, while **Germany achieves 98.4% total return rate** **Error! Reference source not found.Error! Reference source not found..**

Extended Producer Responsibility: Shifting responsibility for product lifecycle management from governments to manufacturers, covering electronics, packaging, batteries, and other products **Error! Reference source not found..**

Green Public Procurement: Government purchasing power leveraged for environmental goals, with **Ireland's public sector representing 10-12% of GDP** **Error! Reference source**

not found. and India's public procurement amounting to 20-22% of GDP Error! Reference source not found..

Market Mechanisms

Voluntary Carbon Markets: Enabling private actors to voluntarily buy and sell carbon credits representing removals or reductions of greenhouse gases Error! Reference source not found.. The market has grown substantially, with both issuance and retirement tripling between 2017 and 2021.

Compliance Carbon Markets: Regulated markets under government programs, including national and subnational emissions trading systems Error! Reference source not found..

Information Instruments

Environmental Labelling: Product labels indicating environmental performance to guide consumer choices toward lower-impact alternatives.

Carbon Footprint Disclosure: Mandatory reporting requirements enabling informed decision-making by investors, consumers, and policymakers.

Innovation and Emerging Instruments

The field continues evolving with innovative mechanisms including:

Nature Performance Bonds: Bonds tied to achieving biodiversity and ecosystem conservation outcomes.

Sustainability-Linked Loans: Financing with terms linked to borrower sustainability performance.

Biodiversity Credits: Market-based mechanisms for financing biodiversity conservation and restoration.

Carbon Border Adjustments: Trade measures to prevent carbon leakage and maintain competitiveness.

4.2.2 CARBON CREDIT-BASED INCENTIVE MECHANISM: A NUMERICAL EVALUATION FOR VIDEO-ON-DEMAND SERVICES USING EUA MARKET PRICING

Integrating carbon credit mechanisms into services such as video streaming (see UC3 and UC4 in EXIGENCE's deliverable D1.2) is expected to provide an economically viable pathway to incentivize sustainable digital behaviour. By aligning user actions with verifiable emission reductions and market-based rewards, EXIGENCE can catalyze a systemic shift toward carbon-aware ICT usage. Figure 12 and Figure 13 show a numerical evaluation for video-on-demand services, assuming a provider with 1 million 4K TV end-user devices (i.e. TV devices capable of 4K video resolution). For example, this is about 1/15 of Deutsche Telecom's Fixed Broadband subscriptions in Germany Error! Reference source not found.. Two incentive mechanisms were considered: (i) direct monetary rewards for individual users (Figure 12.a),

and (ii) a lottery-based scheme (Figure 12.b) aimed at encouraging users to adjust video resolution and switching from TV devices to terminal that consume less energy and quality (Laptop, Tablet, and smartphones). Access to EU Allowance (EUA) market prices via APIs has been found to be technically feasible **Error! Reference source not found.**, offering both hourly resolution and historical data. Additionally, hourly carbon intensity data via API, commercially available **Error! Reference source not found.**, can be integrated into the incentive model, enabling location- and time-specific carbon footprint calculations.

Figure 12 summarizes the results showing that, the most contribution to incentive is provided by the end-user device side representing 93.6% (397,449 €), followed by Video provider CDN with 5.7% (24,341€), and the lowest contribution is obtained from the transmission network (Fixed Broadband) with 0.6% (2,603 €).

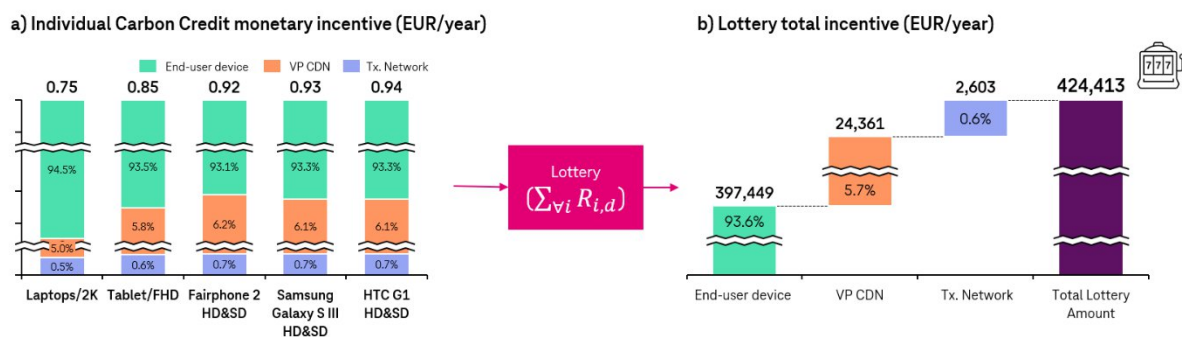


Figure 12: Carbon incentive per user device and network domain contribution.

Figure 13 shows the substantial overall Carbon footprint reduction that can be achieved to be about **6,071 tCO₂e** per year representing about 9.5-fold efficiency gain.

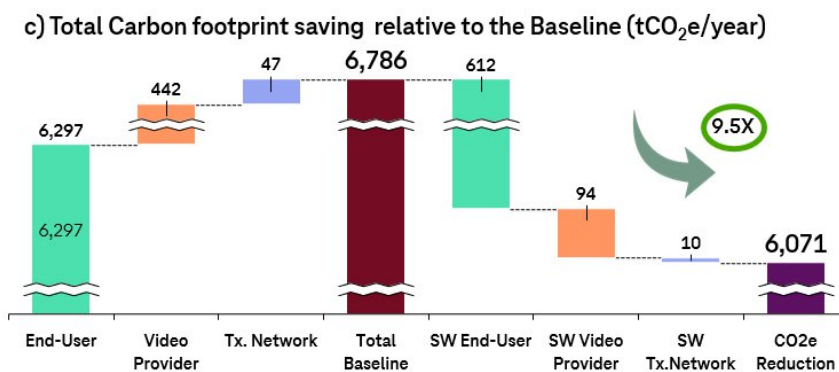


Figure 13: Total Carbon footprint saving relative to the Baseline (tCO₂e/Year)

4.3 OPEN CHALLENGES IN MODELING AND INTEGRATION

At this stage of the EXIGENCE project, we have identified no direct regulatory implications for the proposed incentive mechanisms; nonetheless, several technical and practical challenges remain, as outlined below.

Open Issues for Incentive Mechanisms for Media Streaming and Carbon Footprint Transparency

The current model relies on probabilistic formulas to capture whether end-users adopt incentives. While this provides mathematical tractability, actual user behavior may diverge from modeled probabilities due to context, cognitive biases, or heterogeneous behavior. Crucially, if users misreport or if measurement APIs provide noisy data, the accuracy of the acceptance model decreases.

Moreover, in this use case, the target stakeholders include end-users (both individual and companies), service providers, and potentially regulators. While Table 8 in D3.1 indicates negligible computation and networking, dependencies on API reporting and platform integration are critical. In practice, regulators may require guarantees of fairness in how penalties or rewards are distributed, especially if some users face systematic disadvantages (e.g., fewer green-energy opportunities).

Finally, the effectiveness of our incentives mechanisms depends on parameters characterizing the sensitivity of the user to environmental factors, the level of altruism etc. The value of each such parameter may depend on the gender, demography etc. However, our studies can go further, by providing methods for learning such parameters, and thus leading to personalized incentives rather than to incentives addressing clusters of users.

Open Issues for Incentive Mechanisms for Carbon-Aware AI Services

While the current model provides a rigorous framework for analyzing incentives in carbon-aware federated learning, several open issues remain.

First, the design relies on truthful reporting by end-users. Rational participants aiming to optimize accuracy and energy efficiency may behave adversarially, for example by manipulating participation levels, misreporting data contributions, or deliberately overusing grid energy to distort the AI service provider's penalty mechanism. Such behavior undermines the reliability of energy metrics and distorts incentives.

Second, if storage is minimal, see Table 8 in D3.1, maintaining sufficient historical data for game-theoretical analysis and prediction becomes challenging. This raises the issue of how to design lightweight storage mechanisms, e.g., compressed summaries, that preserve enough history to detect misbehavior or to adapt incentives effectively. Regarding the AI-training in UC10, even though Nash equilibrium is an appealing and universal solution concept in multi-agent strategic interactions, it comes with several issues. One is computational hardness, which we mitigate through heuristics that compute approximate solutions. A second issue is the potential multiplicity of Nash equilibria. Since this is inherent to multi-agent strategic interactions, we inherit it as well. However, if all participants converge to an approximate Nash equilibrium, then approximately neither will deviate from this choice. Thus, multiplicity cares us about the quality of the outcome. We can treat this by examining the different outcomes of the heuristic when it is utilized at various points in the strategic space.

UE: User Incentivization	
Input	User energy data (real-time and historical energy consumption/savings at the end-user device, device type and available actions/decisions)

	Economic data (profit, pricing, subsidies)
	Service usage and API data (received from the measurements)
	Environmental impact metrics
Algorithm	None
Output	Reward/penalty values (e.g., discount, gift values, etc)
	User feedback (real time notification)
	Aggregated reports
Compute	None
Storage	None
Networking	Negligible:
	Data exchange
	Low-latency communication
Energy	Execution: Low, given the low computation, communication, and networking requirements AI-based solutions: High for train ML models

Table 6 User Incentivization

4.4 INTER-DOMAIN AND DEPENDENCIES OF INCENTIVE MECHANISMS

Effective incentive mechanisms depend on having sufficiently detailed and timely measurements. When adequately detailed energy and carbon metrics are available (that is, detailed enough and at the proper timescales), incentives can be accurately computed and properly/fairly attributed, among users and other types of stakeholders, devices etc. If such detailed data are not available, then approximate or aggregated estimates must be used, which may sacrifice accuracy, fairness or automation potential. Therefore, the measurement and data-exchange methods developed in WP2 are essential for enabling credible, transparent, and implementable cross-domain incentive mechanisms.

Table 6 Mapping of WP2 Energy and Carbon Footprint Metrics (EC#, CF#) to Representative Use Cases (UC3, UC4).

Table 7 Energy and Carbon Footprint Metrics Use Cases (UC3, UC4).

Metric	Source (WP2)	Consumer (WP3 mech.)	Required Granularity	Update Frequency	Note
EC#1*	EC of a single user end-to-end session	OTT Service Provider	/Per user session	At session end	Main incentive input metric; captures user-level energy consumption across domains

CF#1	CF of each computation module	Per user session	At session end	Converts EC#1 into user-visible sustainability score or credit
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* If EC#1 is not available, EC#5 can be employed alternatively -loosing some level of granularity- provided that it is calculated on a per user basis. The same applies for EC#11.

Table 7 Mapping of WP2 Energy and Carbon Footprint Metrics (EC#, CF#) to Representative Use Cases (UC10).

Table 8 Mapping of WP2 Energy and Carbon Footprint Metrics (EC#, CF#) to Representative Use Cases (UC10).

Metric	Source (WP2)	Consumer (WP3 mech.)	Required Granularity	Update Frequency	Note
EC#5 **	EC of a complete service monitoring per AI model; AI task execution engine; Aggregated from all nodes and processes	Service host; AI Service Provider;	per service	at job completion	Used to compute end-to-end energy consumption
CF#1	CF of each single traffic flow	Service host; AI Service Provider; Energy Provider	Per user session	At session end	Key metric for carbon-awareness
CF#2, 3	CF of infrastructure HW component/device CF of software components	Service host; AI Service Provider; Energy Provider	Per node and component	Per task or continuously	Can be used for management purposes (e.g. making longer-term modifications of policies)

**If EC#5 is not available, EC#11 can be employed alternatively.

Next we provide clarifications regarding the requirements of the User Incentivization algorithms. First, the algorithms for incentivization run as an offline process, outside

the real-time computational scope of EXIGENCE. While the incentivization logic influences the design of technical algorithms and mechanisms, it does not impose any immediate system-level requirements. Moreover, in terms of algorithmic complexity, the incentivization algorithms are not particularly demanding. Techniques like lottery-based mechanisms involve randomization to allocate rewards, and provide fairness, etc, typically with low complexity. For the “Compute” aspect, there are no requirements for UE within EXIGENCE itself, as the computations are carried out independently, outside the system.

Regarding the distribution of incentive mechanisms across domains such as OTT platforms, ISPs, and end-users, the primary actors involved are the end-user, the Service Provider, and—potentially—the Mobile Network Operator (MNO) and the energy providers. Also, regulators and other relevant policy bodies may play an important role. For example, regarding the media streaming case, as **content providers** encourage users to adopt lower bit-rates or flexible viewing modes, they can tap into energy markets by **partnering with Mobile Network Operators (MNOs) and infrastructure providers (energy providers)**. Reduced energy consumption across the network attained by means of lower bit-rate streaming or off-peak service delivery can result in tangible savings—some of which could be redistributed as shared value. Moreover, by aligning their operations with broader sustainability goals, providers may become eligible for incentives offered by regulators (or other appropriate policy bodies) such as **carbon credits, green financing, and climate-related government subsidies**. These financial mechanisms represent a shift in value creation, where environmental stewardship is not just a cost but an asset. In the following tables, we provide more detailed information about application of the different incentive mechanisms when they span across domains, as also shown in Table 8 and Table

9.

Table 9 Incentive Mechanisms Across 6G Value Network Domains (Media Streaming Case)

Actor	Role in Value Network	Incentive Mechanism	Challenges & Considerations	Potential Benefit
End - User (Individual/Company)	Adopts energy-efficient behaviors (e.g. lower bit-rate, off-peak usage)	Behavioral nudges, environmental points	Cognitive biases, misreporting, heterogeneous preferences; fairness in access to green energy	Potential cost savings, recognition, environmental contribution, personalized incentives

Service Provider	Designs and deploys incentive schemes; interfaces with APIs and platforms	Revenue-sharing, carbon credits, green financing	“noisy” API data/ users misreport, platform integration complexity; need for fairness in reward distribution	Operational efficiency, ESG alignment, monetization of sustainability new revenue streams, enhanced sustainability credentials
Mobile Network Operator (MNO)	Manages traffic and energy load; enables off-peak delivery	Strategic partnerships, shared savings, dynamic contracts	Dependency on accurate API data; integration with OTT and energy providers	Network optimization, new revenue streams, enhanced sustainability credentials
Energy Provider / Infrastructure Operator	Supplies energy; supports demand-response and green energy tracking	Certification schemes, grid balancing incentives, Participation in demand-response programs	Limited visibility of user-level behavior; need of integration with digital platforms	Grid stability, access to climate-related subsidies

Regulators Policy Bodies	/ Set sustainability targets and incentive frameworks	Carbon credits, green subsidies, compliance frameworks	risk that other stakeholders may not comply or align with policy goals ensuring transparency and accountability in the allocation of policy-driven incentives (e.g., carbon credits, green subsidies) issued by the regulator itself.	Policy impact, equitable sustainability outcomes, accelerated adoption of sustainable practices across ICT
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Table 10 Incentive Mechanisms Across 6G Value Network Domains (AI Services)

Actor	Role in Value Network	Incentive Mechanism	Challenges & Considerations	Potential Benefit
E n d - U s e r (Individual Company)	/ Requests AI services via NSP; may offer device resources for distributed training / inference	Behavioral nudges, environmental points, dynamic pricing	Cognitive biases, misreporting, heterogeneous preferences; fairness in access to green energy	Potential cost benefits, recognition, environmental contribution, personalized incentives

AI Service Provider	Owns and orchestrates AI models (training, fine-tuning, inference) across heterogeneous resources (UEs, CDNs, Service Hosts)	Carbon-aware scheduling, carbon-credit schemes, shared-savings with network operators and energy providers	Complex coordination of distributed tasks; dependency on accurate energy/CO2 data	Reduced operational carbon footprint, eligibility for green financing
Network Service Provider (NSP / Operator)	Supports energy- and carbon-aware data transport	Dynamic contracts, shared-savings models, performance-linked carbon credits	Real-time data freshness required; limited granularity in energy reporting from infrastructure; integration with AI Service Provider	Network-wide energy optimization, compliance with sustainability targets
Service Host / Cloud Platform	Manages AI tasks/data exchange, including the necessary connectivity; handles resource allocation	Participation in green-energy markets, pay-for-performance based on carbon efficiency	Workload fluctuation; cooling energy; need for accurate node-level EC/CI metrics	Lower operational costs, increased utilization of renewable energy, carbon-efficiency recognition

Energy Provider / Infrastructure Operator	Supplies energy and carbon-intensity data to NSPs, and Service Hosts; supports demand-response	Dynamic tariffs, renewable-energy certificates, grid-balancing incentives	Need for data interoperability with telecom and cloud APIs; limited granularity of carbon-intensity feeds	Grid stability*
Regulators / Policy Bodies	Sets sustainability targets and incentive frameworks	Carbon credits, green subsidies, compliance frameworks	risk that other stakeholders may not comply or align with policy goals; ensuring transparency and accountability in the allocation of policy-driven incentives (e.g., carbon credits, green subsidies) issued by the regulator itself.	Policy impact, equitable sustainability outcomes, accelerated adoption of sustainable practices across ICT

* Grid stability is identified as a potential benefit for the Energy Provider or Infrastructure Operator. When energy demand and supply are unbalanced, the grid can become unstable, thus potentially leading to power disruptions or even blackouts. In UC10, the proposed incentive mechanisms can encourage end users or service hosts to postpone AI workloads to periods with greater renewable energy availability. This temporal shift in demand helps align energy consumption with green energy supply, thereby supporting grid stability.

5 CONCLUSIONS

In this interim report (MS3.3), we have consolidated the ongoing progress in WP3 and connected it with architectural inputs from WP1 and metric/data contributions from WP2. The documented mechanisms for orchestration, runtime services, and incentives illustrate the technical maturity reached so far, while also clarifying which aspects remain under development. The milestone serves as the bridge between the foundational work in D3.1 and the upcoming deliverable D3.2, aligning WP3 with the overall project objectives.

5.1 PROGRESS SUMMARY

~~WP3 has advanced significantly since D3.1.~~

~~In orchestration, Kubernetes and MANO have been extended with carbon-aware scheduling and placement mechanisms, including the introduction of vEAS and eco-efficiency scoring plugins.~~

~~In runtime services, novel scheduling algorithms and service distribution mechanisms have been defined, tailored for carbon awareness and distributed scalability.~~

~~In incentives, models and simulations have been developed for end-user engagement and provider alignment, with use case-specific adaptations progressing.~~

~~These outputs demonstrate the feasibility of embedding carbon-awareness in system orchestration, execution, and user interaction.~~
Orchestration: Kubernetes and MANO were extended with carbon-aware scheduling and placement mechanisms, validated in a testbed with vEAS and eco-efficiency scoring plugins showing up to 25 % improvement in resource utilization and 20 % reduction in over-provisioning under variable workloads.

Runtime services: Newly implemented scheduling algorithms demonstrated 10–15 % reduction in total energy consumption compared to baseline Kubernetes scheduling in edge-cloud deployments.

Incentives: Simulation models quantified potential user-driven demand reduction of 10–20 %, depending on incentive elasticity and adoption levels.

These quantitative outcomes confirm that carbon-awareness can be practically embedded into orchestration, runtime, and incentive layers, representing clear progress toward the PO2 efficiency targets.”

5.2 OPEN ISSUES AND REMAINING WORK

Despite the progress, several open issues remain:

~~Gross-domain orchestration, multitenant, and agent discovery require further alignment with WP1 architecture and WP2 data collection, and are planned under T3.4 and future deliverables.~~

MANO and Kubernetes integration has achieved intra-domain readiness, but inter-domain applicability remains open.

Runtime mechanisms are designed and validated at small scale; further evaluation under realistic loads is needed to confirm scalability.

Incentive mechanisms are in prototype or model form; full validation with empirical data and user trials will follow in D3.2.¶

Next steps in WP3 focus on consolidation and handover. We will finalize metric-semantic alignment with WP2, harden the vEAS scheduler and eco-control plugins, complete documentation of OSM/Kubernetes interfaces and telemetry endpoints, and package the code for WP4 integration. WP3 will also run lab-scale experiments to calibrate parameters and establish reference baselines and align the EXIGENCE Agent interactions with the WP1 DC functional architecture. These outcomes will be reported on D3.2 and delivered to WP4 for validation.¶

5.3— MILESTONE AND DELIVERABLE LINKAGES¶

~~Despite the progress, several open issues remain:~~¶

~~Gross-domain orchestration, multitenancy, and agent discovery require further alignment with WP1 architecture and WP2 data collection and are planned under T3.4 and future deliverables.~~¶

~~MANO and Kubernetes integration has achieved intra-domain readiness, but inter-domain applicability remains open.~~¶

~~Runtime mechanisms are designed and validated at small scale; further evaluation under realistic loads is needed to confirm scalability.~~¶

~~Incentive mechanisms are in prototype or model form; full validation with empirical data and user trials will follow in D3.2.~~¶

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